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Quality-Driven Next-Best-View Planning for Target-Conditioned Egocentric 3D Reconstruction

Qualitätsgetriebene Next-Best-View-Planung für zielkonditionierte egozentrische 3D-Rekonstruktion

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Transparency in the Use of AI Tools

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Abstract

This thesis studies target-conditioned, quality-driven Next-Best-View (NBV) planning for egocentric 3D reconstruction in Aria Synthetic Environments (ASE). It defines target-specific Relative Reconstruction Improvement (RRI) as an oracle signal, constructs finite candidate and replay contracts, and separates privileged label generation from the actor-visible inputs available to a learned finite-horizon value model. The evaluation is designed to measure oracle-lookahead headroom and the fraction recovered by a learned policy under matched oracle re-evaluation. The evidence available for this version establishes the evaluation contract and implementation readiness, but does not contain confirmatory held-out policy outcomes; it therefore supports no claim that a learned policy improves over the specified baselines. The main remaining limitation is the absence of a validated, population-level rollout bundle with paired endpoint estimates and uncertainty.

Zusammenfassung

Diese Arbeit untersucht zielkonditionierte, qualitätsgetriebene Planung der nächsten besten Ansicht für die egozentrische 3D-Rekonstruktion in ASE. Sie definiert die zielspezifische RRI als Orakelsignal, legt Verträge für endliche Kandidatenmengen und Replay-Daten fest und trennt die privilegierte Erzeugung von Trainingssignalen von den für ein gelerntes Modell mit endlichem Horizont sichtbaren Eingaben. Die Evaluation soll den Spielraum einer vorausschauenden Orakelstrategie und den durch eine gelernte Strategie erreichten Anteil unter identischer Orakel-Neubewertung messen. Die für diese Fassung verfügbare Evidenz belegt den Evaluationsvertrag und die Implementierungsbereitschaft, enthält jedoch keine bestätigenden Ergebnisse auf zurückgehaltenen Daten. Daher wird keine Überlegenheit einer gelernten Strategie gegenüber den festgelegten Baselines behauptet. Die wesentliche verbleibende Einschränkung ist das Fehlen eines validierten Rollout-Datensatzes auf Populationsebene mit gepaarten Endpunktschätzungen und Unsicherheitsangaben.

Contents

Glossary and Abbreviations

Dataset

***ADT* – Aria Digital Twin**: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

***AEO* – Aria Everyday Objects**: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

***ASE* – Aria Synthetic Environments**: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

***CPF* – Central Pupil Frame**: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

***MPS* – Machine Perception Services**: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

***MTD* – Motion Trajectory Data**: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

***MFCD* – Multi-Frame Camera Data**: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

MSDPD **– Multi-Semi-Dense Point Data**: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

Project Aria **– Project Aria**: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

SSL **– SceneScript Language**: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

snippet **– Snippet**: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

VRS **– Virtual Reality Standard**: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

target **– Target of Interest**: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom**: Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum**: Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward**: Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box**: 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom**: Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve**: Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance**: Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \to M}$ and $D_{M \to P}$; older seminar material may still call this CD.

***CR* – Coverage Ratio**: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

***GT* – Ground Truth**: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

***RRI* – Relative Reconstruction Improvement**: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

***reward* – Target-RRI Reward**: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

***target RRI* – Target-Specific RRI**: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

***EFM3D* – Egocentric Foundation Model 3D**: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

***EVL* – Egocentric Voxel Lifting**: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

***Q_H* – Finite-Horizon Q Function**: Target-conditioned finite-candidate value family Q_theta(s,e,a,h) that scores each valid candidate for an explicit requested residual horizon h up to a configured maximum H. The thesis-core architecture is one shared horizon-conditioned scorer; dense Q_1 and exact Q_2 targets establish the recursion, horizon-indexed fitted Q supplies longer-horizon targets, and Double Q remains an optional correction for a noisy learned successor maximum.

***target-conditioned scorer* – Target-Conditioned Scorer**: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

***VIN* – View Introspection Network**: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

***cost* – Acquisition Cost**: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

candidate **– Candidate View**: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

transition **– Counterfactual Transition**: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

action set **– Finite Candidate Action Set**: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

return **– Finite-Horizon Return**: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

5DoF **– Five Degrees of Freedom**: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

CF+ state **– Geometry-Rich Counterfactual State**: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic observations only: rendered depth, valid mask, backprojected point cloud, and derived local normals or support summaries.

historic state **– Historic Snippet State**: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

***CF0 state* – Minimal Counterfactual Actor State**: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

***NBV* – Next-Best-View**: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

***oracle state* – Oracle Rollout State**: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

***offline state* – Persisted Offline Sample State**: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

***state* – Rollout State**: Family of rollout state variants used by ARIA-NBV. The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

***NBV MDP* – Target-Conditioned NBV MDP**: Finite-horizon Markov decision process used to define ARIA-NBV's target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

***mask* – Validity Mask**: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

***GT-EVAL* – Ground-Truth Target Evaluation**: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

***OBS-SEL* – Observed Target Selection**: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

***PRED-Q* – Predicted-Target Q**: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

***MVS* – Multi-view Stereo**: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV's current oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

***PC* – Point Cloud**: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

***SLAM* – Simultaneous Localization and Mapping**: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

***track* – Track**: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

***VIO* – Visual-Inertial Odometry**: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

***Occupancy Grid* – Occupancy Grid**: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

***3DGS* – 3D Gaussian Splatting**: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-

NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

***oracle RRI* – Oracle RRI**: RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
	Oracle and Reconstruction	
RRI	Relative Reconstruction Improvement score for a candidate view.	`oracle.rri`
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	`oracle.points`
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	`oracle.points_q`
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	`oracle.candidates`
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	`oracle.candidates_t`
$q_{t,i}$	Candidate pose i at rollout step t.	`oracle.candidate_qti`
$\boldsymbol{D}_q$	Oracle-rendered or candidate-specific depth map for a proposed view.	`oracle.depth_q`
$D_{P\to M}$	Point-to-mesh directional error component.	`oracle.acc`
$D_{M\to P}$	Mesh-to-point directional error component.	`oracle.comp`
	RRI Metrics	
D	Aggregate point-mesh reconstruction error used by RRI definitions.	`rri.cd_value`
	ASE Assets	
$\mathcal{M}^{\mathrm{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	`ase.mesh`
$\mathcal{M}_e^{\mathrm{GT}}$	Ground-truth mesh crop for target e.	`ase.mesh_target`
$\mathcal{F}^{\mathrm{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	`ase.faces`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(t)$	Project Aria rig pose in the world frame at time t.	`ase.traj`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	`ase.traj_final`
	VIN Scorer	
r	Target or scene RRI value used as a model target.	`vin.rri`
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	`vin.rri_hat`
$\mathcal{L}$	Training loss for scorer or value-model objectives.	`vin.loss`
m	Candidate validity indicator used by scorer diagnostics.	`vin.cand_valid`
	Entity and Target	
RRI_e	Target-specific Relative Reconstruction Improvement for entity e.	`entity.rri_e`

Symbol	Meaning	Key
$\boldsymbol{\varphi}_e$	Actor-visible target/entity descriptor used to condition target-specific value prediction.	`entity.target_desc`
Planning and Rollout		
s	Abstract planning state.	`rl.s`
a	Action index selecting a candidate row.	`rl.a`
r	Scalar reward or immediate gain.	`rl.r`
G	Finite-horizon return accumulated across rollout steps.	`rl.G`
$\mathcal{M}_{\text{NBV}}$	Target-conditioned finite-candidate NBV decision process.	`rl.mdp_nbv`
$\mathcal{A}(s_t)$	Valid action set available in state s_t.	`rl.action_set`
T	State-transition operator for selected candidate actions.	`rl.transition`
s_t^{hist}	Logged historic snippet state at rollout step t.	`rl.s_hist`
s_t^{off}	Persisted offline sample state used by training readers.	`rl.s_off`
s_t^{cf0}	Minimal actor-visible counterfactual state.	`rl.s_cf0`
$s_t^{\text{cf+}}$	Geometry-enriched counterfactual successor state.	`rl.s_cf_geom`
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	`rl.s_oracle`
r_t^e	Target-specific reward at rollout step t.	`rl.reward_target`
$G_t^{(H)}$	H-step finite-horizon return from rollout step t.	`rl.return_h`
Q_H	Finite-horizon candidate-value function.	`rl.qh`
γ	Return discount factor.	`rl.gamma`
H	Planning or rollout horizon length.	`rl.H`
$m_{t,i}$	Hard validity mask for candidate i at rollout step t.	`rl.validity_mask`
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t.	`rl.invalid_reason`
e_t	Selected target entity at rollout step t.	`rl.target`
b_t	Remaining acquisition budget at rollout step t.	`rl.budget`
$C(\tau)$	Acquisition cost of a selected trajectory.	`rl.acquisition_cost`
Shape and Size		
N_q	Number of candidate views in a candidate table.	`shape.Nq`
K	Number of ordinal bins or discrete levels when used in scorer losses.	`shape.K`
scene		
$\boldsymbol{M}_t^{\text{ray}}$	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t.	`scene.ray_memory_t`
Observation and Actor State		

Symbol	Meaning	Key
$\boldsymbol{F}_t^{\text{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	`obs.dino_point_bank_t`
$\boldsymbol{X}_t^{\text{pt}}$	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	`obs.point_tokens_t`
	scene	
$\boldsymbol{\Phi}_t^{\text{scene}}$	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	`scene.scene_memory_t`
$\boldsymbol{E}_0^{\text{EVL-local}}$	Root local EVL evidence field used for target/candidate support reads.	`scene.evl_local`
$\omega_{t,i}^{\text{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	`scene.evl_support_frac`
$\boldsymbol{g}_{t,i}^{\text{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	`scene.evl_support_token`
$\boldsymbol{g}_e^{\text{tgt}}$	Pooled target-support descriptor for the selected target hypothesis.	`scene.target_support_pool`
$\boldsymbol{g}_{t,i}^{\text{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	`scene.frustum_support_pool`
$\boldsymbol{g}_{t,e,i}^{\text{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	`scene.target_frustum_pool`
$\boldsymbol{g}_{t,i}^{\text{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	`scene.ray_query_ti`
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	`scene.render_query`
	spatial	
r_t	Reference pose for candidate-relative descriptor construction at decision step t.	`spatial.ref_pose`
$\boldsymbol{T}_{r_t,i}^{\text{rel}}$	Relative transform from the reference pose r_t to candidate camera i.	`spatial.ref_candidate_transform`
$\boldsymbol{h}_{t,i}^{\text{pose}}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	`spatial.candidate_pose_feat`
$\boldsymbol{h}_{t,e\|i}^{\text{rel}}$	Candidate-target relation descriptor in the candidate/query local frame.	`spatial.candidate_target_rel_feat`
$\boldsymbol{e}_{a\|i}^{\text{rel}}$	Query-local relative positional embedding for target, history, support, or candidate relations.	`spatial.relation_rpe`
	model	

Symbol	Meaning	Key
$\boldsymbol{h}_e^{\mathrm{tgt}}$	Learned selected-target token built from the actor-visible target descriptor and scene support.	`model.target_token`
$\boldsymbol{x}_{t,i}$	Per-candidate row feature assembled from pose, relation, support, validity, provenance, and history descriptors.	`model.candidate_row`

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1 Introduction

Active reconstruction couples sensing and inference: after a fixed acquisition budget, reconstruction quality depends not only on the surface estimator but also on which feasible viewpoints were acquired. Classical active perception therefore treats sensing actions as part of perception, and view-planning work formalizes the recurring generate–score–select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. This thesis studies a deliberately bounded instance of that problem: target-conditioned view selection from a finite admissible candidate table for egocentric indoor reconstruction.

VIN-NBV provides the closest objective precedent by supervising candidate ranking with RRI, computed from the reduction in point–mesh reconstruction error after adding a query view [Fra+25]. ARIA-NBV retains this quality-driven principle but changes the task and planning question. The utility is target-specific rather than scene-wide, and non-myopic value is interpreted only if bounded oracle lookahead improves endpoint quality over one-step selection under the same finite candidate support. Project Aria, ASE, and Egocentric Foundation Model 3D (EFM3D) provide the calibrated egocentric, mesh-supervised, and local actor-visible substrate for this study [Eng+23, Met25a, Str+24].

The current data-generation tasks are defined directly from geometry-valid GT oriented bounding boxes. They provide target identity, target crops, and oracle labels, but they do not implement actor-visible target discovery or identity matching. Consequently, deployable-input claims require a separate observed- or predicted-target protocol; the current target tasks support oracle supervision and controlled evaluation only.

Problem statement. Given a GT-defined target task, actor-visible reconstruction evidence, a finite candidate table with hard validity constraints, and a fixed acquisition budget, determine whether bounded oracle lookahead improves endpoint target reconstruction over one-step oracle-greedy selection and, only if such headroom exists, whether a learned policy using non-privileged inputs recovers a meaningful fraction of it under the same oracle evaluation.

The thesis separates supervision from decision-time information. Ground-truth geometry defines current target tasks, target crops, oracle labels, and evaluation; it is not an ordinary actor input. Invalid candidates are outside the admissible action set rather than examples with low utility. One-step oracle greedy is an immediate-reward comparator over the evaluated valid candidate table, while oracle lookahead is an upper reference only within the fixed candidate generator, horizon, branch factor, target pool, and validity regime. These restrictions make negative results interpretable without turning a bounded experiment into a universal statement about view planning.

1.1 Research Questions

RQ1 — Objective and endpoint contract. Do target-cropped point–mesh error, root-normalized immediate gain, and fixed-budget endpoint gain provide a repeatable objective for comparing target-conditioned view-selection policies?

RQ2 — Offline finite-candidate planning. Under identical target tasks, roots, candidate support, validity rules, and acquisition budgets, does bounded oracle lookahead exhibit positive endpoint-quality headroom over one-step oracle-greedy selection, and can an offline finite-horizon policy recover a meaningful fraction of that headroom?

RQ3 — Actor-visible representation. Which observed or predicted target descriptor and actor-visible state representation are sufficient for learned one-step and finite-horizon candidate scoring without privileged target geometry or all-candidate oracle renders at decision time?

RQ4 — Support and scale. How do target-task coverage, candidate support, hard validity, rollout diversity, and scene-level scale constrain the reliability and generality of the conclusions for RQ1–RQ3?

Together, these questions delimit the evaluated contribution; online control, continuous actions, and real-device deployment remain outside it.

2 Foundations

ARIA-NBV combines ASE trajectories and GT geometry, local EFM3D evidence, oracle RRI labels, and VIN-style finite-candidate scoring. ASE supplies calibrated Aria-like streams and aligned semi-dense maps; EVL supplies frozen local voxel evidence rather than a complete long-horizon scene memory [Met25a, Met25b, Str+24]. GT OBBs currently define data-generation target tasks, whereas observed or predicted target discovery remains outside the implemented actor path.

2.1 Related Work

Classical active perception establishes that sensing actions and inference must be designed jointly, while three-dimensional view planning supplies the finite generate–score–select abstraction used here [AWB88, Baj88, SRR03]. Receding-horizon and submodular formulations explain why lookahead and greedy comparators can be informative under specific utility assumptions [Bir+16, Lau+20]. They do not establish such guarantees for target-specific RRI, whose value depends on the evaluated target, current reconstruction support, and admissible candidates. This thesis therefore tests non-myopic headroom empirically under fixed support rather than assuming it from generic coverage objectives.

Quality-driven NBV replaces geometric coverage proxies with reconstruction-quality supervision. VIN-NBV is the closest precedent because it ranks sampled candidates using oracle RRI rather than visibility alone, while PB-NBV contributes efficient finite projection-based proposals [Fra+25, Jia+25]. Neither establishes target-cropped finite-horizon value under an Aria-native actor/oracle separation. ARIA-NBV consequently uses VIN-style one-step scoring as a myopic control and asks separately whether bounded lookahead improves endpoint target quality.

Target- and object-aware NBV shows why scene-wide utility can conceal the reconstruction needs of one entity. Object-centric NBV motivates conditioning view utility on a designated object, while OANBV and ObjViewBench contribute visibility, feasibility, occlusion, and target-difficulty analyses [Hu+26, Jeo+26, Pan+26]. The current ARIA-NBV data generator does not solve actor-visible target discovery: it samples geometry-valid GT OBB rows as target tasks and evaluates their target crops. Observed or predicted target records are therefore a separate requirement for learned-policy claims, not an implemented consequence of target-specific supervision.

Project Aria defines the calibrated multimodal egocentric sensing regime, ASE supplies synthetic trajectories and GT geometry, and EFM3D/EVL supplies local actor-visible spatial evidence and object hypotheses [Eng+23, Met25a, Str+24]. This stack enables controlled counterfactual labels but does not itself define an NBV objective or make oracle information actor-visible. Meshes, GT boxes, and all-candidate renders remain supervision and evaluation assets; candidate selection must ultimately use logged observations, accumulated selected-view evidence, and an observed or predicted target record.

Offline value learning does not require online environment interaction. Fitted Q iteration estimates a greedy action-value function from a fixed transition batch through a sequence of supervised regression problems [EGW05]. For a bounded variable horizon, fixed-horizon TD defines a family of values in which the horizon-h prediction bootstraps from horizon $h-1$ rather than from itself; the additional functions may share parameters and be updated in parallel [De +20]. Conditioning one shared approximator on the requested horizon follows the more general principle of conditioning value functions on an explicit task variable [Sch+15]. Double DQN is a possible selector/evaluator correction for overestimation in a learned successor maximum, not a prerequisite for offline learning or for the definition of a variable-horizon scorer [HGS15]. CQL and BCQ remain relevant warnings that a max over learned values can select actions whose multi-step consequences are weakly supported by the offline behavior data [FMP19, Kum+20].

Deep Sets and Set Transformer provide permutation-aware baselines for an unordered candidate table [Lee+19, Zah+17]. These methods do not replace the domain contract: invalid actions remain hard-masked, candidate support is reported, and every selected trajectory is re-evaluated by endpoint target quality rather than accepted on predicted value alone.

Geometric representation learning supplies a hierarchy of inductive biases rather than one mandatory backbone. Query-local relative encoding can remove nuisance global coordinates while preserving target, frustum, and motion geometry [Zho+23]. PointNeXt, Point Transformer, KPConv, and sparse convolutions offer coordinate-bearing encoders when explicit point or ray-memory summaries become insufficient [CGS19, Qia+22, Tho+19, Wu+24]. EGNN, SE(3)-Transformer, and GATr provide stronger equivariant alternatives [Bre+23, Fuc+20, SHW21], but exact equivariance is not assumed beneficial for a gravity-aligned, metric, egocentric task; it remains an ablation against simpler local-frame features.

Broader scene memories address a different limitation from candidate interaction. GenNBV distinguishes occupied, free, and unknown grid state, Hestia records directional visibility, SceneScript extracts broad layout and object structure from Aria-style semidense input, and object-aware 3D Gaussian Splatting couples renderable primitives with target membership [Ave+24, Che+24, Jeo+26, Lu+26]. ARIA-NBV uses these works to define representation hypotheses and diagnostics; it does not import their rewards, action spaces, or performance claims into the finite-candidate target-RRI experiment.

2.2 Geometric Learning and Candidate-Set Theory

The learned object in ARIA-NBV is a typed finite-candidate decision map: given a target record, selected history, partial actor-visible geometry, and an unordered table of feasible candidate poses, assign one score to each candidate for selection under later oracle re-evaluation. The model must respect the symmetries and information boundary of this map before architectural capacity is interpreted [Bro+21].

The first required structure is candidate-row permutation equivariance. Reordering $\mathcal{Q}_t$ may reorder per-candidate Q_H outputs, but it must not change the value assigned to the same physical candidate. Deep Sets supplies a pooled symmetry baseline, while Set Transformer supplies candidate interaction without assigning semantic meaning to row order [Lee+19, Zah+17]. Both retain a row-level scoring path because the policy selects a candidate row.

For any permutation matrix Π acting on candidate rows, the value model should satisfy

$$f_\theta(\Pi \boldsymbol{X}_t, \Pi \boldsymbol{m}_t) = \Pi f_\theta(\boldsymbol{X}_t, \boldsymbol{m}_t)$$

where $\boldsymbol{X}_t = \{\boldsymbol{x}_{t,i}\}_{i=1}^{N_q}$ stores per-candidate actor-visible descriptors and $\boldsymbol{m}_t$ stores validity. Invalid and padded rows are outside the admissible action set, not low-utility examples. Selection is therefore

$$\mathcal{A}_t = \{i \in \{1, ..., N_q\} : m_{t,i} = 1\}, \quad a_t^\theta = \operatorname*{argmax}_{i \in \mathcal{A}_t} f_{\theta,i}(\boldsymbol{X}_t, \boldsymbol{m}_t)$$

The second required structure is frame discipline. Candidate, target, current-camera, and history poses use reference- or query-local relative features such as $\boldsymbol{T}_{c_q}^r$ and continuous rotation encodings [Zho+19, Zho+23]. This prevents an arbitrary world origin or row order from becoming a shortcut while preserving gravity alignment and camera-frustum geometry. It is a controlled coordinate choice, not a claim of full global SE(3) invariance.

The third required structure is explicit selected-view history. A camera pose also defines a viewing direction from which target-local surfaces may or may not have been observed. A compact history on $\mathbb{S}^2$ can therefore distinguish directional novelty from generic pose distance [e3n25]. Coverage, overlap, and uncertainty remain diagnostic features unless paired oracle evaluation shows that they improve target-specific RRI [GML22, JLD24].

Object	Required structure	Testable consequence
Candidate rows	Permutation-equivariant per-row value map.	Row-shuffle tests must satisfy $f_{\theta(\Pi X,\Pi m)} = \Pi f_{\theta(X,m)}$ for every selection score.
Invalid and padded rows	Separate feasibility head, trained on every non-padding row; RRI/value loss only on rows certified valid. Hard action masking is the initial policy, with a later calibrated soft feasibility gate.	Invalid rows cannot change valid-row scores except through explicit valid-count or support features.
Candidate-target geometry	Target-local and candidate-local relative pose features.	World-frame origin, yaw convention, and display-only camera transforms cannot become shortcuts.
Selected-view history	Directional memory on $\mathbb{S}^2$.	Novelty tests separate already-observed target directions from generic pose distance.
Actor/oracle boundary	Typed provenance for target descriptors, labels, and support features.	GT-defined tasks, meshes, crops, and all-candidate renders supervise data generation and evaluation only.

Table 1: Minimum geometric-learning contract for the finite-candidate value model.

The implemented hard mask defines $m_{t,i} = 0$ as exclusion from the action set, not as a numerical return. An invalid row's RRI is undefined: it must be neither forced to zero nor assigned a synthetic negative target, because zero can be a legitimate valid-row RRI and either choice would conflate feasibility with quality.

Implementation: planned **Evidence: pending**

A separate feasibility head is planned. It is not part of the current scorer or training module.

Literature: [GE19, Liu+19]

Promotion gate: implement feasibility head and evaluate held-out calibration without weakening the hard mask

Development source: aria_nbv/aria_nbv/rollouts/zarr_store.py; planned finite-horizon scorer

For each non-padding row, the planned model emits a feasibility probability and an RRI/value estimate. Binary cross-entropy supplies negative evidence on invalid rows, while the RRI loss is evaluated only when $m_{t,i} = 1$. This factorization treats feasibility as a reject decision rather than an artificial low-RRI target.

The hard mask remains the deployment safety constraint. A soft feasibility score is only a training and ranking ablation: after held-out feasibility calibration, compare a score proportional to predicted validity times the conditional valid-row value against the hard-mask baseline, while retaining the hard mask wherever it is available. The abstention literature likewise models rejection as a separate outcome rather than a continuous-regression target [Liu+19]. The curriculum hypothesis is therefore to train feasibility from the first batch, warm up the value head only on oracle-valid rows, and then introduce the calibrated soft-gate ablation [Ben+09]. It must never reintroduce an RRI target or RRI gradient for invalid rows.

Research TODO: Compare hard masking against a calibrated feasibility-times-value ranking only after feasibility calibration is measured. The hard mask remains the safety constraint in every comparison.

Source: invalid-row supervision hypothesis

Gate: feasibility calibration and policy ablation

These requirements become replay-field and model acceptance tests in Chapter Section 4.4. Row shuffles must permute outputs, invalid-row perturbations must leave valid scores unchanged, and equivalent local-frame encodings must preserve physical candidate identity. Architectural comparisons are interpretable only after those contracts hold.

The resulting foundation is narrow: active perception motivates action-conditioned sensing, VIN-NBV supplies the quality-driven one-step precedent, ASE and EFM3D define the logged egocentric evidence boundary, and finite-action learning supplies mask and support controls. Coverage and uncertainty remain diagnostics rather than substitutes for target-specific reconstruction quality.

3 Oracle and Data Generation

This chapter defines the non-deployable pipeline that turns logged ASE snippets into supervised target-conditioned NBV tasks. It separates actor state from privileged data-generation state, specifies target-task and candidate construction, defines hard invalidity and target-specific RRI, and records the resulting selected counterfactual chains. The learned method in Chapter 4 may consume only the actor-side projection of these artifacts; GT geometry, counterfactual renders, labels, and oracle search remain instruction, supervision, or evaluation assets.

3.1 State and Visibility Boundary

The learned component in this thesis is a masked finite-horizon candidate-value function: it predicts Q_H values for a finite candidate table, and the decision rule selects among hard-valid rows. Training evidence is generated by offline, mesh-supervised counterfactual replay. ASE contributes both sensor-like logged streams and privileged ground truth; EFM3D converts the logged subset into a local 3D representation, whereas depth, segmentation, GT boxes, meshes, counterfactual renders, and reconstruction labels remain supervision or evaluation channels [Met25a, Str+24]. Storage in the same adapted snippet does not make these channels equally observable.

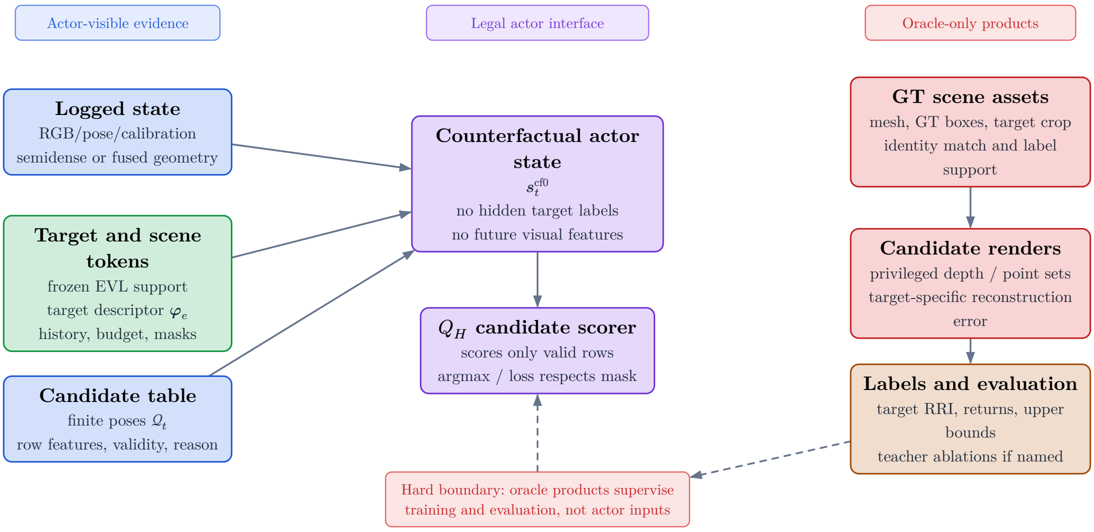


Figure 1: Actor and oracle boundary. Legal Q_H inputs are calibrated logged image evidence, poses, semi-dense geometry with uncertainty and observation support, frozen EVL features or predictions, an explicitly sourced target instruction, selected-view history, remaining budget, candidates, masks, and reason codes. GT depth, segmentation, boxes, meshes, target crops, counterfactual renders, labels, and endpoint evaluation remain privileged.

The visibility boundary is protocol-relative and temporal. A dense render for an unselected candidate at the current decision step is oracle evidence. A render from an already selected action may enter a later state only under an explicitly named source protocol: `CF-GT` for privileged mesh-rendered depth, `CF-sensor` for a declared sensor simulation, or V1 for an actor-visible observation. The same array shape can therefore denote different information, and source role must remain explicit.

3.1.1 ASE/EFM Logged Observation State

The target is treated as external task context, so the value is conceptually $Q(s_t, e, a)$ rather than a target-independent state value. The logged historic state contains the recorded trajectory evidence,

$$s_t^{\mathrm{hist}} = \left(\boldsymbol{I}^{\mathrm{rgb}}, \boldsymbol{X}, \mathcal{P}^{\mathrm{semi}}, \boldsymbol{F}_v, e_t, b_t\right) \in \mathcal{S}^{\mathrm{hist}}$$

Each component of this tuple has a distinct information role:

Calibrated multi-camera image snippets. EFM3D consumes posed and calibrated RGB and greyscale Aria streams. For every stream and time step, EVL encodes the image with a frozen 2D foundation model, projects the centers of a local 3D voxel grid into the distorted camera model, bilinearly samples the feature maps inside the camera's valid radius, and aggregates the samples across cameras and time by their mean and standard deviation [Str+24]. In the repository, `EfmCameraView` binds each normalized frame tensor to `CameraTW` calibration, device-clock timestamp, frame identifier, and an optional separately timestamped distance map. The ASE distance map is GT and remains oracle-only [Met25a]. Thus $\boldsymbol{I}^{\text{rgb}}$ in s_t^{hist} denotes calibrated, time-indexed image evidence; the concrete adapted snippet also carries left and right greyscale SLAM streams rather than a single unposed RGB tensor.

Trajectory, calibration, and gravity. The adapted trajectory is a time-stamped sequence of world-from-rig `PoseTW` transforms with a world-frame gravity vector. EFM3D uses the most recent RGB rig pose, gravity-aligns its rotation, and places a local voxel grid in front of the rig; the per-camera calibration and poses then define the 3D-to-image sampling map [Str+24]. These fields therefore determine the spatial registration, metric scale, gravity direction, and temporal alignment of all lifted evidence. They are also the basis for expressing a candidate relative to the current rig rather than learning values from arbitrary world coordinates.

Semi-dense geometry and observation support. ASE provides MPS-style world-frame semi-dense points, per-point distance and inverse-distance uncertainty, timestamps, and point-to-image observation records [Met25a]. The local `EfmPointsView` preserves padded per-time point slices, valid lengths, metric uncertainty, timestamps, and spatial bounds. EFM3D voxelizes the points into a binary surface mask and uses their observing camera centers to sample the rays before each surface point into a separate free-space mask [Str+24]. Consequently, $\mathcal{P}^{\text{semi}}$ alone denotes sparse surface support; free-space evidence is justified only when observation lineage and camera poses are retained. A missing point is not evidence that a voxel is free.

Derived EVL field and heads. EVL concatenates the lifted image features with the semi-dense surface and free-space masks, processes the volume with a 3D U-Net, and applies task heads for occupancy and 3D OBB detection. The paper's detection head predicts voxel centerness, class scores, and gravity-aligned box geometry; its surface head predicts occupancy and is supervised from GT depth samples labelled as free, surface, or occupied [Str+24]. The resulting $\boldsymbol{F}_v$ is therefore a learned, locally supported statistic of the logged snippet, not another raw sensor modality. The reported EFM3D configuration uses a finite 4 m × 4 m × 4 m grid anchored at the latest rig pose, so an out-of-extent target or candidate has missing EVL support rather than a measured zero or an invalid pose [Str+24].

The remaining tuple elements are decision context rather than ASE/EFM sensing modalities. e_t identifies the entity whose reconstruction is valued, and b_t conditions the return on the remaining acquisition budget. The current V0 target values are derived from a selected GT box; a V1 target must instead come from EVL predictions or another actor-visible association path. This provenance distinction is consistent with `EfmSnippetView`, which exposes camera/calibration, trajectory, and semi-dense fields as actor observations while keeping `ARIA_OBB_PADDED`, nested `gt_data`, and attached ASE meshes on the oracle/evaluation side.

3.1.2 Counterfactual Actor State

The counterfactual actor state retains the immutable logged substrate and adds only information causally available after selected actions,

$$s_t^{\mathrm{cf0}} = \left(\boldsymbol{F}_v^{\mathrm{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t\right) \in \mathcal{S}^{\mathrm{cf0}}$$

For the canonical model, this compact tuple is read together with the explicit ordered selected-view history:

$$s_t^{\mathrm{actor}} = \left(s_t^{\mathrm{cf0}}, \boldsymbol{H}_t\right)$$

In s_t^{cf0}, the root EVL field remains fixed, whereas the accumulated point set $\mathcal{P}_t$ may grow with selected observations. The finite candidate table $\mathcal{Q}_t$ contains poses and generator provenance, not observations from those poses. $m_{t,i}$ defines the admissible action set, $\rho_{t,i}$ records constraint failures, and the ordered history $\boldsymbol{H}_t$ retains the selected approach sequence.

The richer geometry-updated state makes the selected observation channels explicit:

$$s_t^{\text{cf+}} = \left(s_t^{\text{cf0}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{P}^{\text{cf}}, \boldsymbol{n}\right)_{1:t}\right)$$

In the current `CF-GT` protocol, selected $\boldsymbol{D}$ is rendered from $\mathcal{M}^{\text{GT}}$ and paired with a validity mask and candidate-camera calibration. Backprojection yields selected-view surface points $\mathcal{P}^{\text{cf}}$ the corresponding camera-to-surface rays delimit observed free space, and $\boldsymbol{n}$ records local surface orientation. These quantities form a geometry-only successor observation. They do not contain the RGB or greyscale images required by EFM3D's 2D feature encoder and therefore cannot support a fresh EVL field at the selected pose [Str+24].

The transition from t to $t+1$ may use only the observation produced by the selected row a_t. It may fuse its surface and ray evidence and update support, uncertainty, direction, and recency. It may not attach image features, detections, or EVL descriptors unless the protocol also provides the calibrated camera streams from which EFM3D derives them. Counterfactual geometry must therefore retain a source role and explicit absence masks for unavailable image-derived channels.

The counterfactual state is thus an **information state** for decision making, not automatically a complete Markov state of the physical scene. It becomes task-sufficient only if its retained root context, causal dynamic evidence, explicit history, target context, action support, and budget preserve every distinction needed to predict future target-specific return. A pose-only implementation updates $\mathcal{Q}_t$, $\boldsymbol{H}_t$, and b_t without acquiring new scene evidence; it is therefore a deliberately weaker `S0-pose` baseline rather than a realization of the geometry-updated state.

3.1.3 Privileged Oracle State

The oracle state adds complete or counterfactual quantities outside the actor input graph:

$$s_t^{\text{oracle}} = \left(s_t^{\text{cf+}}, \mathcal{M}^{\text{GT}}, \mathcal{M}_e^{\text{GT}}, \boldsymbol{D}_q, \mathcal{P}_q, \text{RRI}\right) \in \mathcal{S}^{\text{oracle}}$$

$$\mathcal{M}_{\text{NBV}} = \left(\mathcal{S}^{\text{hist}}, \mathcal{S}^{\text{cf0}}, \mathcal{S}^{\text{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H\right)$$

ASE provides per-frame metric GT depth aligned with RGB, per-pixel instance identifiers, class mappings, and the scene trajectory; the EFM3D release adds ASE OBB metadata and validation meshes for object-detection and surface-reconstruction supervision [Met25a, Str+24]. EFM3D uses the depth channel to supervise occupancy at sampled free, surface, and behind-surface points, while visible OBBs supervise centerness, class, and box geometry [Str+24]. These uses establish label provenance, not actor observability.

For ARIA-NBV, $\mathcal{M}^{\text{GT}}$ is the privileged reference surface used for candidate rendering and reconstruction evaluation, while $\mathcal{M}_e^{\text{GT}}$ fixes the selected entity's evaluation support. All-candidate depth $\boldsymbol{D}_q$ and backprojected point sets $\mathcal{P}_q$ answer counterfactual queries for unselected rows. RRI then compares target reconstruction error before and after adding that candidate evidence. Depth, points, crops, and RRI are legal for label generation, oracle search, diagnostics, and named upper bounds; none is a contemporaneous input to the student value model.

A GT OBB may define V0 identity, pose, extent, crop, and target-bearing instruction. It does not show that the actor detected or associated that object. V1 must obtain the target hypothesis from EVL predictions or another observation-derived path, retaining ASE identity, segmentation, OBBs, and meshes only for matching and evaluation.

The state hierarchy therefore separates **what was logged**, **what a selected counterfactual history may causally add**, and **what only the oracle can know**. Its layers are related by controlled information projections, not by freely copying arrays between stores. In particular, an oracle tensor may be persisted beside actor tensors for efficient replay without becoming part of the actor state.

3.1.4 Visibility, Feasibility, and Utility

Visibility answers whether a modality contains evidence for a spatial element from a particular view; feasibility answers whether an action is admissible; utility answers how beneficial an admissible action is for the target. These concepts must not be collapsed. A target can be weakly visible from a geometrically valid candidate, a candidate can lie outside EVL support while remaining physically reachable, and an oracle render can fail even though the candidate pose itself is feasible.

Invalidity is a constraint rather than a reward value. Geometry-invalid candidates receive a hard action mask and a persisted candidate reason code before policy selection, stochastic normalization, loss construction, and bootstrap maximization. A geometrically feasible candidate may still have low or negative target gain. Failure of the separate oracle evaluation does not create a candidate reason code: depending on the configured recipe, the affected row or table is skipped, or its `oracle_label_mask` and `q_train_mask` are cleared and the oracle failure is reported separately. Oracle-derived feasibility must be distinguished from a deployable validity estimate whenever policy claims cross from V0 to V1.

3.2 Data Generation and Target-Specific RRI Labels

An oracle target task fixes the entity identity and GT evaluation crop. A deployable target descriptor would instead be predicted from actor-visible observations. The current rollout generator has only the first contract: it projects a selected GT box into a compact instruction for candidate generation. Consequently, the present target descriptor denotes the requested object and its geometry; it does not establish actor-visible target discovery.

The general descriptor notation remains

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\Big)$$

The implemented oracle sampler populates only GT identity, class, confidence, pose, extent, and reference-relative geometry. Projected area, semidense support, EVL support, and proposal-match scores are not measured by this path and must not be interpreted from their schema placeholders.

The finite action interface consists of a candidate table $\mathcal{Q}_t$, hard validity mask $\boldsymbol{m}_t$, and invalid-reason vector $\boldsymbol{\rho}_t$. The admissible set is $\mathcal{A}_t = \{i : m_{t,i} = 1\}$. Invalid rows remain logged for coverage and failure analysis but lie outside policy argmax, sampling, loss targets, and bootstrap maximization. Low RRI is a valid low-utility outcome; it is never an encoding for infeasibility.

For target e, the oracle computes a target-cropped point–mesh error Δ_t^e. Candidate selection and Q_H supervision use root-normalized target gain; state-relative RRI is retained as a diagnostic. Fixed-budget endpoint gain is the intended policy estimand, but the current replay store records cumulative selected-chain gains rather than an independent post-horizon endpoint reconstruction for every policy. Confirmatory policy comparisons must therefore add matched oracle endpoint re-evaluation or an explicitly persisted endpoint record.

3.2.1 Oracle Label Provenance

The one-step scene-level oracle from the seminar work supplies the rendering and point–mesh substrate [Fra+25, Met25c]. The target-specific pipeline renders valid candidates from the GT mesh, backprojects their depth, fuses candidate points with the privileged current evaluation cloud, crops both points and mesh to the selected target box, and evaluates the resulting point–mesh error. In the current protocol the root evaluation cloud is reconstructed from ASE GT depth, so both the crop and the root metric are oracle-only.

Camera geometry is therefore useful here only as part of a relation: a logged trajectory, a target crop, a finite candidate set, or a rendered oracle query. The calibrated camera boundary is not a contribution by itself. The consolidated candidate-support figure in Figure 3 uses camera glyphs only to expose those relations; native `CameraTW` unprojection and world-from-camera `PoseTW` remain the geometry owners behind the render path.

3.2.2 Target Selection

The implemented sampler does not match actor proposals to GT objects. It enumerates the non-padding GT OBB rows in a snippet, accepts rows with finite positive geometry, and applies seeded uniform sampling without replacement up to the configured per-snippet cap. Thus the stored `matched` status currently means geometry-valid GT task row, not successful proposal-to-identity association. IoU, ambiguity-gap, visibility, and support thresholds are absent from this admission rule.

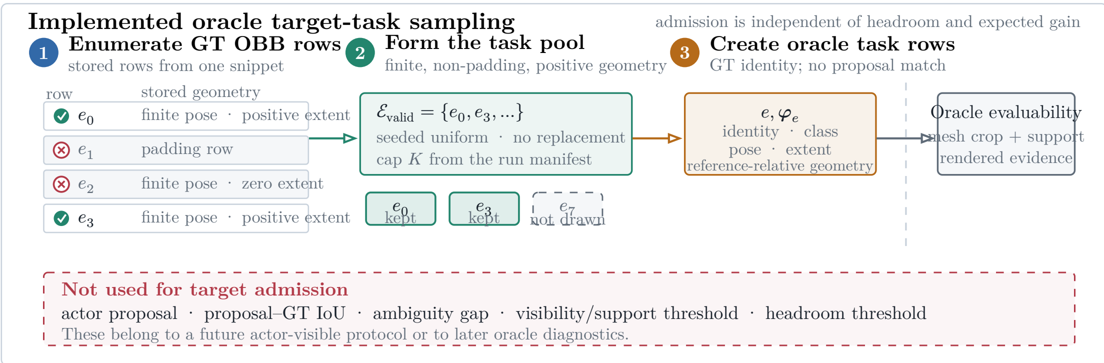


Figure 2: Implemented oracle target-task sampler. Geometry-valid GT OBB rows form the task pool, and seeded uniform sampling without replacement applies the manifest-defined cap. Rollout scoring later decides whether the selected crop is evaluable. No actor proposal, IoU match, visibility gate, or support gate is used.

The sampler bounds rollout cost without pre-filtering on headroom. Candidate scoring may subsequently invalidate the task when its mesh crop, current support, or rendered evidence is unusable; otherwise near-solved and negative-gain targets remain scientifically informative. A later actor-visible protocol must introduce proposal identity and observation-quality diagnostics as a separate selection stage rather than retroactively interpreting the present oracle fields as measurements.

3.2.3 Candidate View Generation

Candidate generation turns one oracle instruction into a finite action table. Every quantitative choice—source population, target cap, shell size, family weights, motion limits, pruning thresholds, rollout recipes, renderer settings, and retention policy—belongs to the resolved run manifest and report bundle. The Methods text defines semantics only; it does not designate one mutable TOML profile as canonical.

At rollout step t, candidate generation constructs a full shell

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = 60$$

with a fixed provenance component $k(i)$ per row. The core mixture contains forward-local, target-bearing, and lateral-bypass motion; the diversity challenger may additionally allocate mass to local refinement and revisit/backtrack components. The resolved manifest, rather than prose, determines which components and counts apply to a run.

For each row, the raw direction is sampled in the reference rig frame. The realistic profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

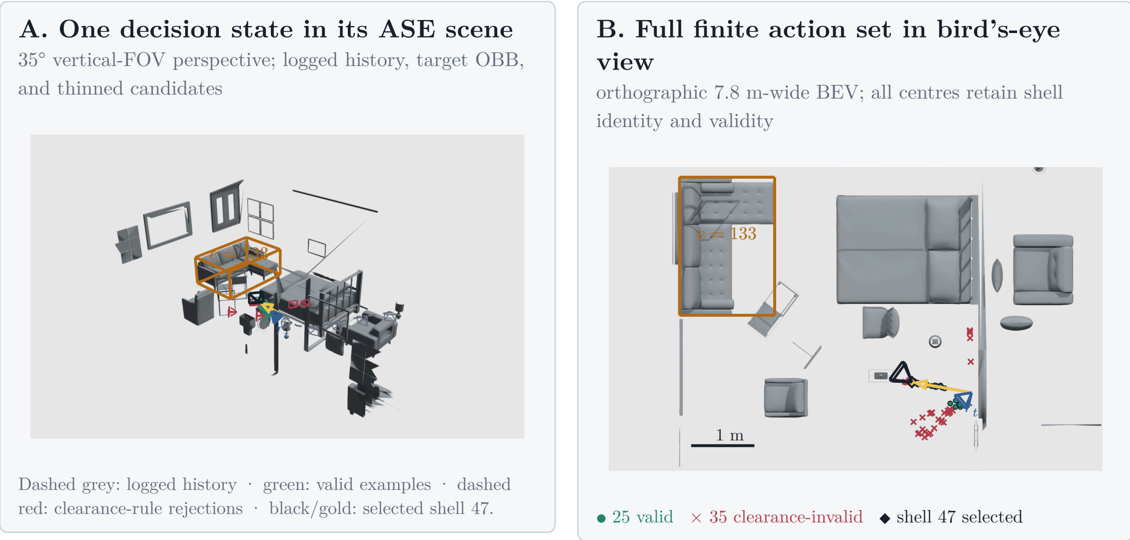
$$\boldsymbol{u}_i \sim \mathrm{PS}(\boldsymbol{e}_z, \kappa), \quad p(\boldsymbol{u}) = c_\kappa (1 + \boldsymbol{e}_z^T \boldsymbol{u})^\kappa, \quad \kappa = 8$$

The draw is mapped into configured azimuth and elevation caps without rejection. With $\psi = \operatorname{atan2}(u_x, u_z)$ and $u_y = \sin\theta$, the cap transform is

$$\psi' = \psi \frac{\Delta_\psi}{2\pi}, \quad y' = \sin\theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin\theta_{\max} - \sin\theta_{\min})$$

$$\boldsymbol{d}_i^0 = \left\| \left(\sqrt{1 - y'^2} \sin\psi', y', \sqrt{1 - y'^2} \cos\psi' \right) \right\|$$

The sampler draws a capped direction in the reference rig frame and reinterprets it as egocentric forward motion, target-bearing motion, lateral bypass, local refinement, or backtracking according to component provenance. Target-looking families orient their optical axis toward the oracle instruction. In this chapter that point is privileged; calling the generator target-conditioned does not make the point actor-visible.



Scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, step row 121 24 forward-local + 24 target-bearing
1. 12 lateral-bypass candidates. Dense mesh layers are

z-buffered raster; OBBs, paths, centres, and wire frusta remain vector geometry.

Figure 3: One pinned finite-candidate decision state from ASE scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, and step row 121. Panel A uses a 35-degree vertical-FOV perspective view to place logged camera history, root state, target OBB, selected path, and a deterministically thinned set of wire frusta in the real scene. Panel B uses a 7.8-metre-wide orthographic bird's-eye view and retains all 60 candidate centres: 25 rows are admissible, 35 are hard-rejected by the clearance rule, and oracle-greedy selects shell 47. Dense scene geometry is z-buffered; OBBs, paths, centres, and camera glyphs remain vector overlays. Full eye, look-at, up, clipping, and resolution parameters are recorded in the figure JSON. This is an auditable contract example, not a policy-performance result.

The three core position families then reinterpret this capped direction. Let $\boldsymbol{f} = \boldsymbol{e}_z$ be the rig-forward unit vector, $\boldsymbol{b}_e$ the supplied target bearing in the reference frame, $\boldsymbol{l}_e = \|\boldsymbol{e}_y \times \boldsymbol{b}_e\|$ the horizontal lateral direction, and $\boldsymbol{e}_y$ the world-up direction expressed in the sampling frame. The family directions are:

$$\boldsymbol{d}_i^{\text{forward}} = \left\|\boldsymbol{f} + \alpha_f(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{f})\boldsymbol{f})\right\|, \quad \alpha_f = 0.45$$

$$\boldsymbol{d}_i^{\text{target}} = \left\|\boldsymbol{b}_e + \alpha_t(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{b}_e)\boldsymbol{b}_e)\right\|, \quad \alpha_t = 0.4$$

$$\boldsymbol{d}_i^{\text{bypass}} = \left\|0.55\boldsymbol{b}_e + 0.85\,\text{sign}(d_{i,x}^0)\boldsymbol{l}_e + \text{clip}\left(d_{i,y}^0, -0.35, 0.35\right)\boldsymbol{e}_y\right\|$$

Finally, the sampler draws a radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}(0.25, 1.1), \quad \boldsymbol{c}_i^w = \boldsymbol{T}_r^w\left(r_i \boldsymbol{d}_i^{k(i)}\right)$$

`forward_local` keeps the reference rig orientation. Target-looking families orient the camera toward the supplied target center $\boldsymbol{p}_e$:

$$\boldsymbol{z}_i^w = \|\boldsymbol{p}_e - \boldsymbol{c}_i^w\|, \quad \boldsymbol{y}_i^w = \left\|\boldsymbol{e}_y - \left(\boldsymbol{e}_y^T \boldsymbol{z}_i^w\right)\boldsymbol{z}_i^w\right\|, \quad \boldsymbol{x}_i^w = \boldsymbol{y}_i^w \times \boldsymbol{z}_i^w$$

These equations describe the sampler, not an optimal proposal distribution. `forward_local` preserves egocentric continuity, `target_bearing_local` moves along the target ray, and `lateral_target_bypass` introduces side-step views; the optional challenger families test smaller corrections and reversals. Candidate-profile utility must be judged after pruning. A store in which target-aware families rarely survive cannot support a target-conditioned planning claim.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\boldsymbol{o}_i\|_2 \le 1.0 \text{ m}, \quad |\Delta h_i| \le 0.25 \text{ m}, \quad \max\left(0, -o_{i,z}\right) \le 0.25 \text{ m}, \quad \Delta\psi_i \le 70 \text{ deg}$$

The full shell is retained with position, strategy, mixture, sampling probability, rule masks, diagnostics, and invalid-reason bitsets. Panel B of Figure 3 makes the distinction concrete for one stored table: invalid rows remain inspectable but sit outside the admissible set. Invalid candidates cannot enter Q_H selection, stochastic normalization, or loss targets. Conversely, a feasible row with weak target support or low expected gain remains a valid low-utility example rather than receiving an invalid-reason code. A manifest-defined root-support threshold may reject an entire rollout task when too few actions remain:

$$N_{\text{valid}} \ge \max\left(12, \text{ceil}\left(0.25 N_q\right)\right)$$

This threshold is a data-support guard, not a reward threshold. Preflight reporting must retain rejected-root counts and failure reasons by candidate family.

3.2.4 Rollout Branch Sampling and Dataset Impact

Rollout recipes select and retain finite chains from the valid action table. The implemented families are uniform valid sampling, one-step oracle greedy selection, bounded oracle lookahead, and temperature-softmax sampling. Their horizon, branch factor, beam width, temperature, and seed are resolved parameters. These recipes generate replay diversity and bounded references; they do not constitute a learned policy.

Bounded oracle lookahead can select a different first action from one-step greedy because it ranks retained finite-horizon chains rather than immediate gain alone (Figure 4). The persisted artifact contains these selected or beam-retained chains and their full per-step candidate shells; it is not an exhaustive materialization of the counterfactual action tree.

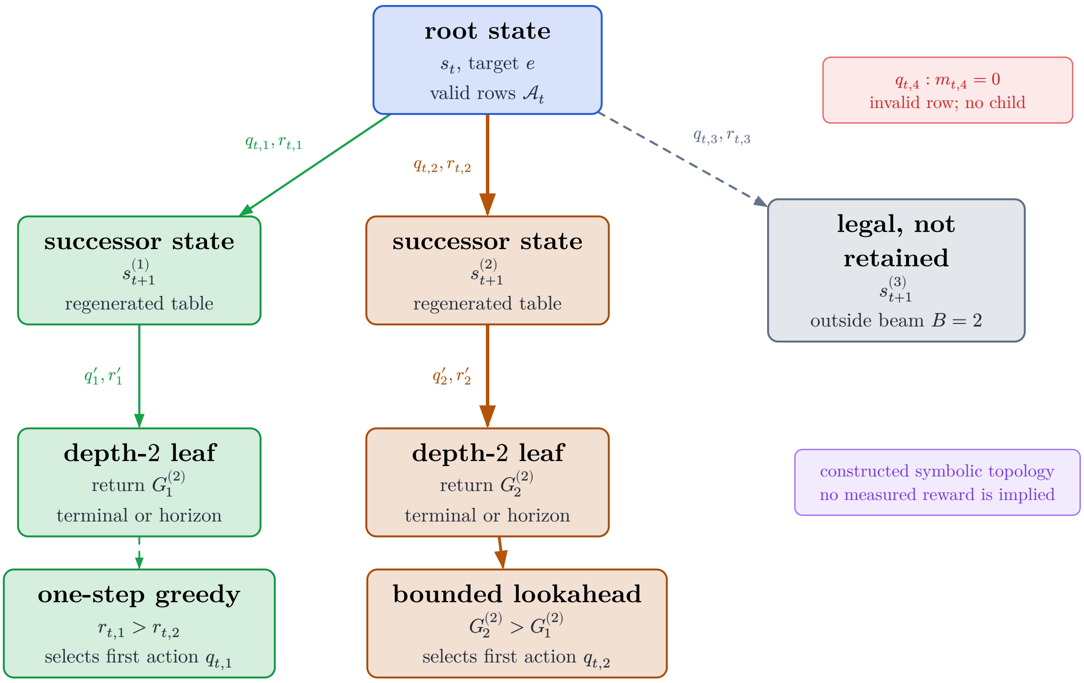


Figure 4: Constructed symbolic topology of the bounded oracle-lookahead reference. Nodes are counterfactual states and edge labels are candidate actions with immediate rewards. Only valid first-action rows produce children; the beam retains a bounded prefix set, and invalid rows receive no branch. The inequalities illustrate how the selected first action can differ from one-step greedy without inventing measured reward values.

For stochastic branches, let s_i be the finite oracle score of valid row i. The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \mathrm{median}(s)}{\mathrm{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1}} \exp\left(\ell_j\right)$$

The target source defines the supervised task, the candidate mixture defines learnable action support, validity defines admissibility, and the recipe defines which chains enter replay. Reporting must therefore cover target-task coverage, valid fanout and invalid reasons by family, selected-family diversity, gain distributions, and retention cost before attributing policy behavior to non-myopic planning. The paired train-only pilots are bandwidth and candidate-profile probes; they are non-confirmatory and provide no held-out policy-performance evidence.

3.2.5 Target-Specific RRI

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated evaluation points to the selected target region. The target error adapts the VIN-NBV objective to this crop [Fra+25]: point-to-mesh accuracy plus mesh-to-point completeness.

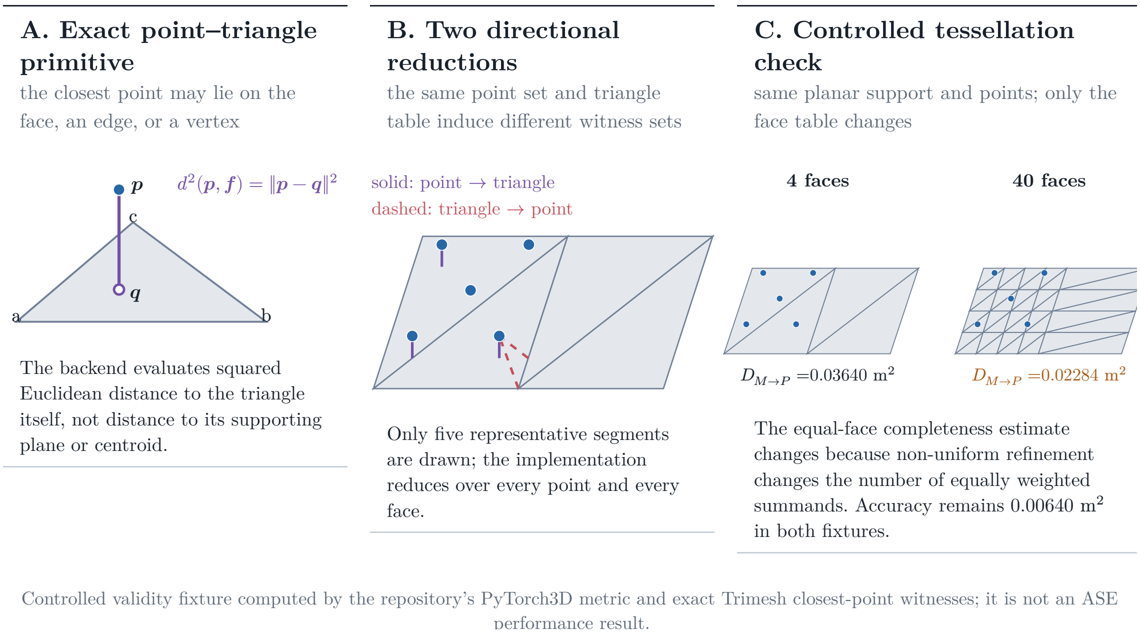


Figure 5: Point–mesh metric contract and controlled validity fixture. Panel A isolates the exact point-to-triangle primitive. Panel B distinguishes the point-to-face accuracy and face-to-point completeness reductions using computed closest-point witnesses. Panel C holds the planar support and point set fixed while changing only the triangle table; the measured equal-face completeness term changes from 0.03640 to 0.02284 m^2. These values are generated by the repository's PyTorch3D metric on a synthetic fixture and demonstrate a tessellation-sensitivity mechanism; they are not an ASE performance result.

$$\Delta_t^e = d\big(C_e(\mathcal{P}_t), \mathcal{M}_e^{\mathrm{GT}}\big) = D^e_{P\to M,t} + D^e_{M\to P,t}$$

The implementation crops mesh faces when any vertex lies inside the oriented target OBB and evaluates the configured point–mesh scorer. Geometry-invalid candidates are removed by the hard action mask and retain persisted candidate reason codes. Empty mesh crops, insufficient current support, or unusable renders instead invalidate the separate oracle evaluation: the recipe either skips the affected row or table, or clears `oracle_label_mask` and `q_train_mask`, while reporting the oracle failure independently rather than assigning a candidate reason code. The controlled fixture in Figure 5 demonstrates that the current equal-face completeness reduction is not invariant to non-uniform retessellation, even when the planar support and point set are fixed. Its sensitivity to root and candidate point sampling remains unmeasured. Confirmatory use therefore requires the sensitivity tests and metric-validity gate specified in Chapter 5; freezing one mesh does not by itself establish a representation-independent oracle.

The immediate training reward adapts VIN-NBV's reconstruction-improvement idea to a target crop and normalizes by the root target error rather than the current error [Fra+25]. This makes equal-horizon rollouts additive against a common root baseline:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_0^e + \varepsilon}$$

The finite-horizon return is the discounted sum of those target rewards along a selected counterfactual branch. It is a training target for Q_H, not a claim that the deployed system has an online continuous-control policy:

$$G_{t,e}^{(h)} = \sum_{k=0}^{\min(h,b_t)-1} \gamma^k r_{t+k}^e$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The log-gain variant is retained only as an internal scale-sensitivity ablation:

$$J_{e,\log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

$J_e^{(H)}$ is the fixed-budget evaluation metric, $G_t^{(H)}$ is the learning return, and $J_{e,\log}^{(H)}$ is a sensitivity diagnostic. Root-normalized gains telescope to endpoint gain when the discount is unity and geometry, horizon, and acquisition budget are matched. State-relative one-step RRI remains a VIN-compatible diagnostic. The resolved manifest must freeze the discount, clipping, target cap, crop policy, and all evaluation-geometry parameters for each reported experiment.

3.3 Replay Stores and Diagnostic Evidence

The pipeline materializes two stores with different ownership. Immutable `vin_offline/` caches logged snippet evidence and expensive one-step oracle products. Standalone `rollouts.zarr/` references those source rows and stores target tasks, retained rollout chains, per-step finite candidate shells, masks, reason codes, selected-action successor evidence, and a derived Q_H view. This separation prevents counterfactual experiments from mutating the source substrate and prevents one-step candidate labels from being mistaken for multi-step replay.

The source store is immutable because it defines the logged actor substrate. Its manifest records source identity, split, schema, materialized blocks, and provenance. A source row may contain frozen EVL/VIN fields, one-step candidate poses and labels, compact rendering payloads, and semidense geometry/history. These fields support scorer training and rollout construction, but they do not themselves define target-conditioned finite-horizon data. Exact directory names, array groups, chunking, and codec choices are versioned reproducibility metadata rather than scientific entities, so they are not reproduced as directory-tree figures in the main text.

The rollout store is normalized around replay identity. One source can produce multiple target rows, each target can produce multiple recipe and retained-chain rows, each chain expands into selected steps, and each step owns a full candidate shell. The store therefore captures selected or beam-retained chain evidence, not the complete counterfactual search tree. Padded `q_h/` arrays are a validated cache over the factual row tables. Invalid candidate rows remain inspectable, but action, training, and bootstrap masks exclude them.

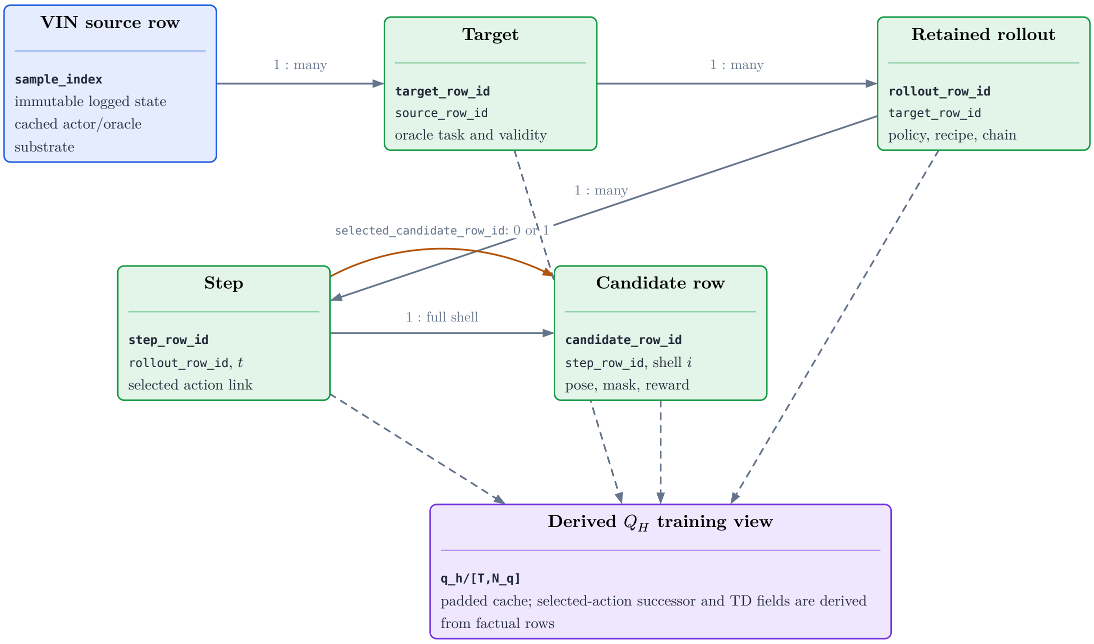


Figure 6: Normalized lineage of the implemented replay evidence. An immutable VIN source row may define several oracle target tasks; each target may produce several retained policy chains; each chain contains ordered steps; and each step owns one full candidate shell. The step row may identify one chosen candidate through `selected_candidate_row_id`; selected-action successor and TD fields are then constructed in the derived `q_h/` join rather than persisted as a separate transition table.

Evidence family	Interpretation contract
Sources and targets	Manifest-backed task coverage; not proof of actor-visible target discovery.
Candidates and invalidity	Full-shell support with separate hard-action, training, padding, and future deployable-feasibility roles.
Retained chains and steps	Recipe-selected evidence; not a persisted exhaustive search tree.
Selected depth	Chosen-action successor observation with calibration and source role; actor input only under an explicitly admitted later-state protocol.
Q_H view	Derived training cache whose rewards and masks must agree with factual rows; not a scene-memory representation.

Table 2: Interpretation contract for rollout-store audits. Numeric values are rendered from the resolved report bundle in the experiment and reproducibility sections.

Selected-depth persistence stores only the depth raster and calibration for the chosen action at each retained step. It is sufficient to reconstruct the selected-observation prefix without duplicating dense all-candidate renders, but persistence does not decide visibility. A `CF-GT` reader may use previously selected GT-mesh depths to build a privileged dynamic state; a deployable reader must instead consume a declared sensor-like or observed source. The current unselected candidate renders remain oracle-only in every student protocol. Selected depth is also not an independently scored endpoint artifact.

Likewise, rollout rows summarize final cumulative selected-chain metrics; they do not preserve every rejected branch or a policy-neutral endpoint reconstruction. These limitations must be resolved by matched endpoint re-evaluation before confirmatory policy comparison.

Scientific reporting is generated from the same inspection frames used by the diagnostic application. The main text needs target-task coverage, candidate validity and invalid reasons, family survival and selection, selected-history sanity, gain distributions, source-role counts, and runtime/storage summaries. Exact schema columns, compression, chunking, hashes, and cluster invocation belong in the reproducibility appendix and must be read from the resolved manifest and report bundle. Development bandwidth pilots are train-only feasibility checks; their counts and throughput may size later jobs but cannot support held-out reconstruction or policy claims.

4 Method

ARIA-NBV is formulated as target-conditioned selection from a finite candidate table. The implemented substrate generates and evaluates masked multi-step pose rollouts, records selected transitions, exposes a dense training view, and provides scorer-independent fitted Double-Q optimization for an injected candidate-value model. It does not yet provide a production Q_H scorer, checkpoint, or policy result. The existing myopic VIN scorer remains the historical one-step control, while the target-conditioned scene-memory scorer described in this chapter remains planned.

The current canonical direction is a bounded fixed-horizon candidate scorer whose actor state includes remaining budget. An explicit requested-horizon query $Q_{\theta(s_t,e,i,h)}$ is a design candidate to compare before the scorer PR, not an implemented or frozen interface. Dense one-step values, exact horizon-two targets, recursive fitted values, Double-Q backups, and behavior-policy returns remain distinct potential objectives or controls; the first scorer design and objective sequence must be fixed by the source-owner decision gate.

The chapter first fixes the actor-visible state and the coverage limitation of local EFM3D evidence. It then separates implemented replay/training infrastructure from proposed scorer DTOs, finite-action semantics, geometric acceptance tests, scene-carrier alternatives, interaction architectures, and bounded Q_H learning targets. Descriptive status blocks distinguish implementation maturity from evidence maturity throughout.

4.1 Actor State and Representation Boundary

4.1.1 Implemented carrier and information boundary

Implementation: partial **Evidence: pending**

The replay carrier, actor/supervision separation, and scorer-independent fitted-Q adapter are implemented. No production finite-horizon scorer or task-sufficient dynamic reconstruction memory is implemented, and frozen scientific validation remains pending.

Literature: [Str+24]

Promotion gate: retain actor/oracle provenance checks and name the admitted state protocol in every run

Development source: aria_nbv/aria_nbv/data_handling/qh.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/tests/lightning/test_qh_module.py

The implemented one-step scorer consumes an actor-visible snippet view, a row-aligned candidate table, a reference rig pose, calibrated candidate cameras, and optionally cached EVL output. Oracle RRI labels enter the loss after prediction and are not scorer inputs. The V0 target descriptor contains semantic identity, pose, positive metric extents, and reference-relative pose in an actor-safe shape, but its current values are derived from privileged GT target tasks. A deployable claim therefore still requires observed or predicted target fields with explicit provenance.

The implemented `QhActorTensors` contract can carry root semidense evidence, GT-derived V0 target geometry, root-relative candidate poses, selected-pose history, and remaining budget to an injected scorer. This is an information boundary, not an implemented model architecture. Candidate regeneration changes pose, history, budget, and the finite action table; a future scorer must still define how causal selected observations update its scene representation.

Three boundaries are invariant. Invalidity is a hard mask with versioned reason codes, not a small value. GT meshes, target crops, current all-candidate renders, associations, and returns remain oracle or audit fields. Previously selected GT-mesh depth may enter a later state only under an explicitly privileged `CF-GT` protocol; the corresponding sensor-like or observed variants must use distinct provenance. Candidate rows remain tied to stable shell identities and documented coordinate frames.

4.1.2 Task-sufficient actor state

Implementation: planned **Evidence: pending**

The canonical state separates immutable root context from a causal dynamic memory. Particular scene carriers are promoted only when they preserve the information boundary and improve target-conditioned decisions.

Literature: [Bro+21, Zah+17]

Promotion gate: selected-observation reader, deterministic fusion, source-dropout tests, and held-out target-RRI ranking

Development source: docs/contents/theory/efm3d_scene_embeddings.qmd; docs/contents/theory/candidate_view_dependence.qmd

ARIA-NBV does not require one universal reconstruction format. It requires an actor-visible state from which a model can compare a target e and candidate $q_{t,i}$ at decision step t. The following information contract includes the requested residual horizon h only for the explicit-horizon design candidate; the current fixed-horizon direction instead derives time-to-go from the bounded task and remaining budget:

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\left(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\right)$$

For architectural and DTO purposes, the scene state is decomposed as

$$\Phi_t^{\mathrm{scene}} = \left(\Phi_0^{\mathrm{static}}, M_t^{\mathrm{dynamic}}\right)$$

where Φ_0^{static} contains immutable logged evidence such as root semidense geometry and supported local EVL features, while M_t^{dynamic} contains only evidence causally produced by selected observations, support/free/unknown state, recency, and directional history. The target is external task context in $Q(s_t, e, a)$ and is represented by a separate `TargetState`; it is not duplicated inside target-independent scene memory.

The selected-pose history $\boldsymbol{H}_t$ remains explicit unless a promoted memory is demonstrated to be a sufficient statistic for it. Raw selected depth is an observation consumed by the memory update; it need not remain a direct scorer input once its surface, free-space, support, source, and recency information have been fused.

State protocol	Dynamic evidence	Interpretation
`S0-pose`	selected poses only	Planned first scorer baseline over the implemented replay tensors; not a complete reconstruction state.
`S1-points`	causally fused selected surface points	Minimum geometry-updated counterfactual state; does not distinguish observed free from unknown space.
`S2-ray`	surface, free, unknown, support, recency	Canonical planned dynamic state for candidate-frustum and target-support queries.
`CF-GT` / `CF-sensor` / V1	source tag on selected observations	Orthogonal information protocol: privileged GT depth, declared sensor-like simulation, or actor-visible observation.

Table 3: Counterfactual state and source protocols. Scene carrier, information source, interaction architecture, and learning objective are orthogonal experimental choices.

A useful representation must preserve distinctions that can change target-specific return, distinguish missing evidence from predicted free space, and admit a causal update after selection. Candidate queries should read the same memory in target-local, candidate-local, and ray-relative coordinates. A common world translation or yaw convention must not change physical candidate values, while gravity, scale, target extent, camera direction, occlusion, and temporal order remain observable task variables. Exact global SE(3) invariance is therefore neither required nor claimed.

4.1.3 Local EFM3D evidence and the coverage gap

Implementation: partial **Evidence: pending**

EFM3D fields, their voxel pose, and finite extent are persisted. No finite-horizon scorer currently consumes them, and their sufficiency as the only long-horizon scene representation remains pending.

Literature: [Met25b, Str+24]

Promotion gate: report target and candidate support against the persisted voxel pose and extent

Development source: docs/literature/tex-src/arXiv-EFM3D/method.tex, Sec. Egocentric Voxel Lifting, lines 2–44; docs/contents/literature/efm3d.qmd; docs/contents/theory/efm3d_scene_embeddings.qmd; aria_nbv/aria_nbv/data_handling/offline/writer.py

EFM3D encodes logged RGB frames before lifting selected evidence into a finite gravity-aligned voxel field anchored to a snippet pose. The stored voxel transform and extent define the support of lifted features and dense heads. They must not be interpreted as a global volume containing every region observed by the trajectory.

A target or candidate outside this support can remain physically valid; it has missing local-EVL evidence rather than an ordinary zero feature or automatically invalid action.

The coverage limitation motivates a layered interface rather than repeated inference at hypothetical candidate poses. Local EVL features provide high-quality Aria-native evidence where supported. Broader semidense or fused point carriers can preserve observed surfaces beyond that volume. Sparse ray-aware state can distinguish occupied, free, and unknown regions and can be updated from selected geometry. Logged appearance descriptors can be attached to points only after visibility and source lineage are established. None of these carriers creates RGB, DINO, detector, or EVL evidence at an unvisited counterfactual pose.

4.1.4 Ranked representation design space

Implementation: exploratory **Evidence: pending**

This table ranks scene carriers only. The interaction architecture is specified separately in Section 4.4 so a carrier change does not silently redefine candidate or target semantics.

Literature: [Ave+24, Che+24, Fra+25, Jeo+26]

Promotion gate: promote only after matched support, leakage, runtime, and oracle-policy ablations

Development source: docs/literature/tex-src/arXiv-GenNBV/3-Method.tex, Secs. Formulation and Generalizable State Embedding, lines 10–49; docs/literature/tex-src/arXiv-Hestia/sec/3_method.tex, Sec. Methods, lines 14–58; docs/literature/tex-src/arXiv-scene-script/sections/structured_scene_language.tex; docs/contents/literature/efm3d.qmd

Carrier	Status	Inductive benefit	Cost and promotion gate
Root semidense moments	planned first scorer carrier	Cheap root-level context for contract and optimization tests.	No spatial support or causal update; never interpret as task-sufficient state.
Persisted EVL and snippet evidence	available root carrier	Aria-native local features, OBB hypotheses, calibrated cameras, and explicit extent.	Not consumed by a finite-horizon scorer; limited spatial support and requires coverage metadata.
Semidense or fused point memory	first dynamic control	Observed surfaces follow trajectory and selected-view extent.	No free/unknown state; test density and visibility sensitivity.
Sparse ray-aware memory	planned primary extension	Separates surface, free, unknown, support, uncertainty, and causal updates.	Requires deterministic fusion and counterfactual-source masks.
Target-centred EVL re-lifting	adaptation ablation	Reuses logged frame features when the target lies outside the root field.	Domain shift in the 3D neck; compare against simple logged-feature pooling.
DINO-on-point	appearance ablation	Extends logged appearance to observed points beyond the EVL grid.	Needs visibility gating, compression, and missing-descriptor masks.
TSDF/SDF or sparse encoder	geometry/encoder ablation	Compact metric geometry or learned coordinate-bearing tokens.	Must preserve observation weights and unknown-space semantics.
Object-aware 3DGS	renderable-memory ablation	Candidate rendering, soft target membership, and primitive uncertainty.	Per-scene optimization and mask supervision change the state contract.
SceneScript	global-context control	Broad ASE-aligned layout and object hypotheses from semidense input.	Not an ATEK drop-in and does not provide causal free/unknown updates.

Table 4: Ranked scene-carrier design space. Status describes the role in the thesis method, not empirical superiority.

Representation comparisons begin with the smallest planned scorer carrier and then add the smallest carrier that tests a diagnosed information loss. Learned point, sparse, equivariant, or renderable encoders are introduced only when a simpler carrier demonstrably limits target-RRI ranking or finite-horizon recovery. This ordering keeps representation capacity separate from target identity, candidate support, replay validity, and the candidate interaction architecture.

4.2 Descriptor and Encoding Protocol

4.2.1 Persisted replay interface

Implementation: implemented **Evidence: pending**

The factual replay schema, lazy reader, framework-neutral dataset seam, and dense `q_h/` view are implemented and unit-tested. Frozen scientific replay evidence remains pending. The schema is a storage contract; readability from the store does not make a field actor-visible.

Promotion gate: preserve row identity, masks, provenance, and source roles in every tensor reader

Development source: aria_nbv/aria_nbv/rollouts/zarr_store.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/aria_nbv/data_handling/qh.py; aria_nbv/tests/rollouts/test_qh_reader.py

The rollout store preserves source, target, rollout, step, candidate, diagnostic, and lineage tables. Its derived `q_h/` arrays provide a padded state–candidate view without changing factual identities or labels. Target pose and extents in the V0 task remain privileged GT-derived instructions. Candidate geometry, selected history, remaining budget, and hard masks are actor-side fields in the implemented DTO seam; target gains, GT associations, mesh diagnostics, crops, and current all-candidate renders remain label or audit fields. No production finite-horizon scorer currently consumes these tensors. Previously selected depth becomes a later actor input only under a named counterfactual-observation protocol.

Carrier	Persisted content	Learning role
Target	identity, class, pose, extents, reference-relative pose, source and validity provenance	privileged V0 task instruction; learned actors need observed or predicted equivalents
Candidate	stable row identities, world/root-relative pose, masks, reasons, sampler provenance, support fields	finite action row; privileged diagnostics remain source-gated
Selected chain	selected row, shell index, step order, policy, seed, successor link, terminal state	history and temporal-difference linkage
Selected observation	selected depth, valid mask, calibration, pose and source	later dynamic-state input only under `CF-GT`, `CF-sensor`, or V1; never an all-candidate student input
Oracle labels	target RRI, target root gain, errors, optional crops and candidate renders	supervision, evaluation, and audit only

Table 5: Implemented replay carriers and admissible learning roles.

4.2.2 Planned model-input and DTO contract

Implementation: partial **Evidence: pending**

The implemented DTO seam separates actor inputs, selected-transition supervision, and audit lineage for varying stored chain lengths. The production scorer DTO remains planned. Static scene context, dynamic selected-observation state, target state, and candidate rows are common requirements; an explicit requested-horizon value query is an alternative pending the source-owner decision.

Literature: [De +20, Bro+21, Sch+15, Zho+23]

Promotion gate: fixed-H versus requested-horizon source-owner decision, typed selected-observation state, positive-width V1 target path, source masks, and leakage tests

Development source: aria_nbv/aria_nbv/data_handling/qh.py; aria_nbv/aria_nbv/lightning/qh_module.py; docs/contents/theory/candidate_view_dependence.qmd

The intended input for target e at step t is

$$\mathcal{I}_{t,e} = \left(\boldsymbol{h}_e^{\mathrm{tgt}}, \boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{H}_t, \boldsymbol{b}_t, t, H, \left\{\boldsymbol{x}_{t,i}, \boldsymbol{e}_{a|i}^{\mathrm{rel}}, m_{t,i}, \boldsymbol{\rho}_{t,i}\right\}_{i=1}^{N_q}\right)$$

This equation is an information contract rather than one flat tensor. The corresponding DTO roles are:

DTO role	Candidate content	Visibility
`StaticSceneContext`	root semidense evidence, supported EVL tokens, root frame and EVL extent	actor input; immutable within one rollout
`DynamicSceneState`	selected geometry, free/unknown support, recency, source masks and ordered history	actor input; causal update only
`TargetState`	protocol-specific descriptor, target-local support and field-availability masks	V0 instruction or V1 actor-visible target
`CandidateTable`	row identity, local pose, target relation, actor validity and padding mask	actor input; row-aligned
`ValueQuery`	optional requested residual horizon h and its availability mask	proposed model-visible query if the explicit-horizon design is selected
`CandidateSupervision`	one-step root gain, diagnostic target RRI and `q_train_mask`	supervision only
`SelectedTransition`	factual action index and row id, reward, discount, terminal and successor identity	training linkage only
`AuditLineage`	source/store/config hashes, policy, seed and reason vocabulary	CPU audit data; not a learned feature

Table 6: Production-scorer DTO design space. Model inputs, optional horizon queries, supervision, transition linkage, and provenance remain distinct even when collated in one training batch.

The maximum supported horizon H is an experiment contract and becomes a checkpoint contract once a scorer exists. In the current fixed-horizon direction, remaining budget b_t belongs to the rollout state and supplies time-to-go context without a separate requested-horizon input. The alternative explicit-horizon design would add h with $1 \leq h \leq b_t \leq H$ so one scorer can answer several residual-horizon queries. That alternative, separate $Q_1, ..., Q_H$ heads, and fixed-H models must be compared before the public scorer interface is frozen. The step index t remains lineage by default and becomes a learned feature only in a named non-stationarity ablation.

The common model boundary is conceptually `score(static_context, dynamic_state, target_state, candidate_table, time_context)`, where `time_context` is remaining budget for fixed-H and may additionally contain requested horizon if that design is selected. The explicit-horizon candidate could evaluate one state for several residual horizons and return $\left[B, L, N_q\right]$, but this is proposed rather than implemented. Either interface must keep static encodings reusable and must not expose future observations.

Padding masks, modality-presence masks, source-role masks, action masks, and training masks remain distinct. If the explicit-horizon design is selected, its horizon-availability mask is also distinct and an unsupported horizon must not be silently clamped to the available budget. A missing modality must never be encoded as an ordinary zero observation or confused with padding.

The target descriptor separates identity, geometry, observed support, confidence, and source:

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\Big)$$

The learned token combines that descriptor with target-local scene support:

$$\boldsymbol{h}_e^{\mathrm{tgt}} = \mathrm{MLP}_{\mathrm{tgt}}(\mathrm{concat}(\boldsymbol{\varphi}_e, \boldsymbol{g}_e^{\mathrm{tgt}}))$$

In the current V0 data, target geometry is GT-derived and several generic descriptor fields are unmeasured placeholders. The model contract should therefore use protocol-specific variants such as `V0GtTargetState` and `V1ObservedTargetState`, or carry an explicit availability mask per optional field. The same numerical zero cannot simultaneously mean absent support and measured zero support.

Target-independent static scene encodings may be reused across several target tasks. The candidate table, however, is not generally shared: target-bearing, lateral-bypass, and target-looking candidates depend on the selected target. Multi-target evaluation must therefore either carry one candidate table per target or construct a union table with a target–candidate availability mask; only physically identical rows may share candidate encodings.

4.2.3 Relative candidate geometry

Implementation: partial **Evidence: pending**

Root-relative candidate geometry and a minimal candidate-local target relation are implemented. The complete candidate-frustum and dynamic-memory relation descriptor remains planned.

Literature: [Li+21, Zho+19, Zho+23]

Promotion gate: frame-transform, row-shuffle, and held-out descriptor ablations

Development source: aria_nbv/aria_nbv/data_handling/qh.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/vin/modules/pooling.py

Canonical rigid transforms remain in the store. The reader or tensor adapter derives a candidate pose relative to the current decision reference rather than learning the arbitrary world origin:

$$\boldsymbol{T}^{\mathrm{rel}}_{r_t,i} = \boldsymbol{T}^{-1}_{w,r_t}\boldsymbol{T}_{w,c_{t,i}}, \quad \boldsymbol{\delta}^{p}_{r_t,i} = \boldsymbol{R}^{\top}_{r_t}\left(\boldsymbol{c}_{t,i} - \boldsymbol{c}_{r_t}\right), \quad \boldsymbol{R}_{r_t,i} = \boldsymbol{R}^{\top}_{r_t}\boldsymbol{R}_{t,i}$$

$$\boldsymbol{h}^{\mathrm{pose}}_{t,i}\left(q_{t,i}; r_t\right) = \mathrm{concat}\left(\boldsymbol{\delta}^{p}_{r_t,i}, \mathrm{R6D}\left(\boldsymbol{R}_{r_t,i}\right), \left\|\boldsymbol{\delta}^{p}_{r_t,i}\right\|_2, \mathrm{atan2}\left(\delta^{y}_{r_t,i}, \delta^{x}_{r_t,i}\right), \Delta h_{t,i}, \boldsymbol{u}^{\mathrm{up/frustum}}_{t,i}\right)$$

The target relation is encoded separately so candidate self-motion cannot be confused with target conditioning:

$$\boldsymbol{h}^{\mathrm{rel}}_{t,e|i}\left(q_{t,i}, e\right) = \mathrm{concat}\left(\boldsymbol{\delta}^{p}_{e|i}, \left\|\boldsymbol{\delta}^{p}_{e|i}\right\|_2, \cos\theta^{\mathrm{opt}}_{t,e,i}, \beta^{\mathrm{elev}}_{t,e,i}, \lambda^{\mathrm{obb}}_{t,e,i}\right)$$

Continuous rotation features, metric range, bearing, elevation, height, and frustum variables preserve physical task structure. Query-local relative positional encoding is a later ablation for candidate–target, candidate–history, or candidate–candidate interactions:

$$\boldsymbol{\delta}^{\mathrm{p}}_{a|i} = \boldsymbol{R}_{t,i}^{\top}(\boldsymbol{p}_a - \boldsymbol{c}_{t,i}), \quad \boldsymbol{\delta}^{\mathrm{R}}_{a|i} = \boldsymbol{R}_{t,i}^{\top}\boldsymbol{R}_a$$

$$\boldsymbol{\eta}_{a|i} = \mathrm{concat}\left(\boldsymbol{\delta}^{\mathrm{p}}_{a|i}, \left\|\boldsymbol{\delta}^{\mathrm{p}}_{a|i}\right\|_2, \mathrm{enc}_R\left(\boldsymbol{\delta}^{\mathrm{R}}_{a|i}\right)\right), \quad \boldsymbol{e}^{\mathrm{rel}}_{a|i} = \psi_{\mathrm{rel}}\left(\mathcal{F}\left(\boldsymbol{\eta}_{a|i}\right)\right), \quad a \in \{j, e, k\}$$

This adopts QCNet's query-centric geometric discipline, not its trajectory decoder or streaming claim [Zho+23]. World-frame copies remain audit facts; the learned interface should expose one authoritative local transform rather than recomputing the same relation through multiple fields.

4.2.4 Candidate rows and directional history

Implementation: partial **Evidence: pending**

The implemented DTO carries set-valued selected-pose history, but no production scorer consumes it yet. Ordered temporal encoding and directional memory remain candidate requirements for longer-horizon models.

Literature: [e3n25, Lu+26]

Promotion gate: directional-novelty fixture and matched architecture ablation

Development source: aria_nbv/aria_nbv/data_handling/qh.py; docs/contents/theory/candidate_view_dependence.qmd

The proposed candidate row is a local query, not a duplicate of the full state. It contains candidate self pose, candidate–target relation, and candidate-local scene/support reads. Shared target, scene, ordered history, and remaining-budget context are supplied once as state tokens; a requested-horizon token is added only if that design is selected. Hard action validity and padding remain masks; candidate family or sampler provenance is audit-only by default and may enter the model only as a named ablation.

Viewing history is not reducible to camera distance. For a target-local point or cell, selected camera directions define a signed first moment together with the second-moment memory

$$\boldsymbol{\mu}^{\mathrm{dir}}_t(\boldsymbol{v}) = \frac{\sum_{k<t} w_k(\boldsymbol{v})\boldsymbol{d}_k(\boldsymbol{v})}{\sum_{k<t} w_k(\boldsymbol{v}) + \varepsilon}, \quad \boldsymbol{M}^{\mathrm{dir}}(\boldsymbol{v}) = \frac{\sum_{k<t} w_k(\boldsymbol{v})\boldsymbol{d}_k(\boldsymbol{v})(\boldsymbol{d}_k(\boldsymbol{v}))^{\top}}{\sum_{k<t} w_k(\boldsymbol{v}) + \varepsilon}$$

The signed first moment distinguishes opposite approach directions, whereas the second moment records concentration along viewing axes. Their query-relative projections form directional features without committing to a full spherical-harmonic field. Low-order spherical harmonics are promoted only if these moments improve over ordered pose history and still leave systematic directional errors.

Optional appearance, ray, and directional blocks require three independent indicators where applicable: modality presence, batch padding, and evidence source. Removing one optional source must alter only its masked branch, and counterfactual-only geometry must not receive fabricated RGB, DINO, detector, or EVL descriptors.

4.3 Finite Candidate and Replay Contract

Implementation: implemented **Evidence: pending**

Finite candidate tables, hard masks, lineage, selected transitions, selected-depth persistence, and the derived `q_h/` view are implemented and schema-tested. Frozen scientific store evidence remains pending.

Promotion gate: preserve deterministic shell identity, source roles, and selected-transition validation

Development source: aria_nbv/aria_nbv/rollouts/replay/types.py; aria_nbv/aria_nbv/rollouts/replay/engine.py; aria_nbv/aria_nbv/rollouts/zarr_store.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/tests/rollouts/test_qh_reader.py

At step t, candidate generation returns a finite full-shell table $\mathcal{Q}_t$ with a hard-valid mask $\boldsymbol{m}_t$ and versioned invalid-reason bitsets. Scores are stored compactly only for hard-valid rows and are bound back to stable shell indices before selection. The admissible action set is

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \{i \in \{1, \ldots, N_q\} : m_{t,i} = 1\}$$

Invalid rows remain available for diagnostics and dense replay, but cannot be selected. A row enters the training mask only when it is actor-selectable and has a finite oracle target. Invalid rows have false masks and undefined labels; scene RRI is never substituted for a missing target-specific label. `valid_action_mask`, `q_train_mask`, padding, and any deployable feasibility estimate are distinct fields with distinct owners.

4.3.1 Implemented replay transition

Rollout expansion records the full candidate table, selected valid and shell indices, policy scores and probabilities, selection policy, and random seed. The implemented transition is

$$(x_{t+1}, \boldsymbol{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \mathrm{Step}\Big(x_t, \boldsymbol{H}_t, b_t, q_{t,a_t}, \xi_t\Big)$$

where x_t is the current reference pose, $\boldsymbol{H}_t$ the selected-pose history, b_t the remaining budget, and ξ_t the deterministic generation context. The next candidate table is regenerated around the selected pose under the same target task, history constraints, and versioned generator configuration.

This transition is deliberately a replay-control transition, not yet a complete reconstruction-state update. It changes pose, selected-pose history, budget, lineage, and action support. It does not imply that the actor has received a new RGB observation, recomputed EFM3D field, or fused the selected depth into a spatial memory. Any implementation that consumes only these fields must be labelled `S0-pose`.

The persisted factual tables retain source and target identity, lineage hashes, step order, selected rows, masks, reasons, sampler provenance, rewards, selected-depth calibration, and support diagnostics. The derived `q_h/` view right-pads states, stores one-step and target-root-gain labels, and exposes only the factual selected transition needed for a temporal-difference backup. It does not create counterfactual labels for unselected actions or make privileged selected depth actor-visible.

4.3.2 Selected observation and planned scene-state transition

Implementation: planned **Evidence: pending**

A task-sufficient successor state must update only evidence produced by the selected observation. Current GT-mesh selected depth is privileged counterfactual evidence and requires an explicit `CF-GT` state protocol.

Literature: [Che+24, Lu+26]

Promotion gate: typed selected-observation reader, deterministic fusion, source masks, and no-future-observation tests

Development source: docs/contents/theory/efm3d_scene_embeddings.qmd; aria_nbv/aria_nbv/oracle/evidence.py; aria_nbv/aria_nbv/rollouts/zarr_store.py

A selected observation is the typed tuple

$$o_{t+1}^{\mathrm{sel}} = \big(D_{t,a_t}, V_{t,a_t}, K_{t,a_t}, T_{\mathrm{root} \leftarrow \mathrm{cam},t,a_t}, \ell_{t,a_t}^{\mathrm{src}}\big)$$

containing depth, valid mask, calibration, root-relative camera pose, and a source role. The source role distinguishes privileged GT-mesh depth, declared sensor-like simulation, and an actor-visible sensor observation. Unselected candidate renders at step t are never elements of the student state.

The existing geometry-level counterfactual uses a set union of retained points,

$$\mathcal{P}_{t+1} = \mathcal{P}_t \cup \mathcal{P}_{q_t}$$

but a planning memory must additionally preserve whether space is observed occupied, observed free, or still unknown. The proposed sparse update is

$$\boldsymbol{M}_{t+1}^{\mathrm{ray}} = \mathrm{Fuse}\big(\boldsymbol{M}_t^{\mathrm{ray}}, \boldsymbol{P}_{t+1}^{\text{selected-obs}}(a_t), \mathcal{R}_{t+1}^{\text{selected-obs}}(a_t)\big), \quad a_t \in \mathcal{A}_t$$

Here both evidence terms are indexed by the selected action a_t; no unselected candidate render enters the successor state. The update may add selected surface evidence, carve observed free space, update support and uncertainty, and refresh target-local directional memory. It must not attach RGB, DINO, detector, or EVL descriptors to counterfactual geometry unless a corresponding actor-visible observation exists. Counterfactual-only cells instead carry explicit source and missing-modality masks.

Raw selected depth is an input to the state builder, not necessarily a permanent model token. The reader may emit either the typed selected-observation prefix or a deterministically derived `DynamicSceneState`, but those two representations must not be mixed silently. This split prevents two common errors: treating pose-only replay as complete visual simulation, and treating privileged selected-depth geometry as a deployable sensor stream.

4.4 Geometric and Mask Acceptance Tests

Implementation: partial **Evidence: pending**

The replay and fitted-Q infrastructure covers row-aligned masks, local-frame tensors, selected-transition admission, and deterministic Double-Q selection for an injected scorer. No candidate-to-state architecture or scorer-level equivariance contract is implemented.

Literature: [De +20, Bro+21, Lee+19, Zah+17]

Promotion gate: production scorer plus end-to-end permutation, mask, duplicate, frame, source, and horizon tests for every admitted state protocol

Development source: aria_nbv/aria_nbv/data_handling/qh.py; aria_nbv/tests/lightning/test_qh_module.py; aria_nbv/tests/rollouts/test_qh_reader.py

Candidate order carries no task meaning. For a per-candidate scorer f_θ, jointly permuting row-aligned inputs by Π must permute outputs by the same amount:

$$f_\theta(\Pi \boldsymbol{X}_t, \Pi \boldsymbol{m}_t) = \Pi f_\theta(\boldsymbol{X}_t, \boldsymbol{m}_t)$$

Equivariance alone does not guarantee invalid-row isolation. Holding valid rows and the mask fixed while changing only invalid-row contents must satisfy

$$\mathrm{Mask}(\boldsymbol{X}_t, \boldsymbol{m}_t) = \mathrm{Mask}(\boldsymbol{X}'_t, \boldsymbol{m}_t) \quad \Rightarrow \quad \mathrm{Mask}(f_\theta(\boldsymbol{X}_t, \boldsymbol{m}_t), \boldsymbol{m}_t) = \mathrm{Mask}(f_\theta(\boldsymbol{X}'_t, \boldsymbol{m}_t), \boldsymbol{m}_t)$$

Mask ownership depends on the interaction. In the candidate-to-state design candidate, candidates are queries and scene, target, history, and time context are keys and values. The action mask therefore sanitizes candidate queries, gates output selection, and gates supervised losses; it is not a key-padding mask for the shared state tokens. Candidate masks become attention-key masks only in candidate-as-key architectures such as DeepSets context or a masked Set Transformer. Padding masks, action-validity masks, training masks, and modality-presence masks must remain separate; a horizon-availability mask is added only if explicit horizon queries are selected.

Duplicate-row and valid-count tests are required because candidate-set pooling or per-set normalization can otherwise change the absolute value of an unchanged physical candidate. A duplicate row may duplicate an output, but it must not silently change another row's value unless the tested architecture explicitly models candidate-set context.

Coordinate handling is deliberately weaker than exact SE(3) equivariance. World poses remain available for reproducibility, while model inputs use root-, target-, or candidate-relative geometry. Gravity, scale, height, yaw, camera direction, target orientation, motion limits, and frustum geometry remain physical variables. A global origin convention must not become a shortcut [Bro+21].

Actor/oracle provenance is the final acceptance condition. Target gains, GT associations, mesh distances, current candidate renders, and target crops may supervise or audit a model but may not enter its actor graph. Previously selected GT depth may enter only a `CF-GT` state branch. Source-dropout tests must show that removing an unavailable optional carrier changes only its masked branch.

If the explicit requested-horizon interface is selected, it adds its own acceptance contract: the boundary query $h = 0$ has value zero; an $h = 1$ target contains no bootstrap; $h > b_t$ is rejected or masked rather than clamped; and a target for $h > 1$ may depend only on a successor value requested at $h - 1$ [De +20]. Vectorizing a horizon axis must reproduce independent calls for every admissible h without exposing future observations. The fixed-H alternative instead requires separate tests across configured horizons and remaining budgets. Neither design assumes monotonicity such as $Q_{h+1} \geq Q_h$ because valid actions may have negative immediate gain.

4.4.1 Orthogonal architecture ladder

Implementation: partial **Evidence: pending**

No level in the interaction ladder is implemented for finite-horizon scoring. Scene-carrier choice, fixed-H versus requested-horizon conditioning, and candidate interaction are orthogonal decisions to test after the scorer boundary is frozen.

Literature: [Bre+23, Fuc+20, Lee+19, SHW21, Sch+15, Zah+17, Zho+23]

Promotion gate: freeze the scorer time-query contract, then promote a level only after lower interaction controls pass on the same scene carrier and target/source protocol

Development source: aria_nbv/aria_nbv/data_handling/qh.py; docs/contents/theory/candidate_view_dependence.qmd

The proposed A1 query contains candidate-local pose, candidate–target relation, and candidate-local support reads. Target, scene, ordered history, and remaining-budget context are supplied once as shared state tokens. The requested-horizon design candidate additionally embeds h; the fixed-H alternative does not. Candidate provenance and generator-family identity are audit-only by default. Shared conditional and separate-horizon models remain alternatives for the source-owner decision rather than a frozen architecture [Sch+15].

Level	Interaction model	Scientific role
A0	Independent masked MLP	Locks target/candidate descriptors, masks, time context, and label learnability for a fixed scene carrier.
A1	Candidate-to-state cross-attention	Each candidate query independently reads shared target, scene, history, and budget context.
A2	DeepSets candidate context	Tests whether an unordered summary of valid candidates adds information without pairwise attention.
A3	Masked Set Transformer	Tests candidate–candidate interaction while preserving row equivariance and mask isolation.
A4	Query-local relation bias	Tests QCNet-style target, history, and candidate relations without importing its forecasting decoder.
A5	Temporal/recurrent state read	Tests whether ordered long-horizon history remains informative after explicit dynamic scene memory.
A6	Residual value head	Tests finite-horizon recovery over a calibrated continuous one-step root-gain control.
A7+	Exact-equivariant or graph interaction	Escalates only after local-frame controls reveal a symmetry-related failure.

Table 7: Planned interaction-architecture ladder. Scene carrier, target/source protocol, time-query contract, and learning target are frozen independently for each comparison.

Candidate-to-candidate attention is not required by the core task. It may improve relative policy context or diversity, but unrelated sampled rows must not silently redefine the absolute value of candidate $q_{t,i}$. Exact equivariant layers are similarly scoped to diagnosed support encoders or candidate graphs after local-frame scalar controls. This ordering favors reusable context encodings and keeps optional requested-horizon conditioning separate from candidate-set interaction.

4.5 Target-Conditioned Finite-Horizon Value Model

4.5.1 Implemented training infrastructure and planned scorer

Implementation: partial **Evidence: pending**

The one-step VIN/CORAL scorer remains the historical myopic control. Replay-chain tensors and a selected-transition Double-Q trainer for an injected scorer are implemented. No deterministic H=2, candidate-to-state, or other production finite-horizon scorer is implemented.

Literature: [CMR19, Fra+25, HGS15]

Promotion gate: retain the one-step scorer as a matched control and implement the first finite-horizon scorer only after its interface decision

Development source: aria_nbv/aria_nbv/vin/models/target_myopic.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/data_handling/qh.py

The implemented actor DTO can carry root semidense evidence, GT-derived target pose and extent, root-relative candidate geometry, selected-pose history, and remaining budget. It deliberately excludes supervision and audit lineage from scorer inputs. These tensors make an `S0-pose` baseline possible, but they do not establish an architecture, checkpoint, or task-sufficient reconstruction state.

Any injected scorer must emit one continuous value per candidate row, while the Lightning adapter owns hard-mask admission, Double-Q target construction, loss, and target synchronization. Candidate-to-state attention, candidate–candidate interaction, and the representation carrier remain production-scorer design choices.

4.5.2 Requested-horizon scorer design candidate

Implementation: planned **Evidence: pending**

An explicit requested-horizon query is one candidate design for sharing scene, target, and candidate encoders across horizons. The current canonical direction remains fixed-H with remaining budget in state until the source-owner gate selects and documents an interface.

Literature: [De +20, EGW05, Lee+19, Sch+15, Zah+17, Zho+23]

Promotion gate: fixed-H versus requested-horizon source-owner decision, dynamic-state reader, A0/A1 controls, horizon tests, and held-out oracle policy comparison

Development source: docs/typst/thesis/sections/04-method/04-01-scene-representation-requirements.typ; docs/typst/thesis/sections/04-method/04-02-descriptor-and-encoding-plan.typ; aria_nbv/aria_nbv/lightning/qh_module.py

For target e, the return over exactly the requested residual horizon h is

$$G_{t,e}^{(h)} = \sum_{k=0}^{\min(h,b_t)-1} \gamma^k r_{t+k}^e$$

and the learned quantity is

$$Q_{h,e}(s_t, i) = \mathbb{E}\left[G_{t,e}^{(h)} \mid s_t, a_t = i\right], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t, \quad Q_{0,e}(s, i) = 0$$

The notation Q_H names a bounded finite-horizon family. If the explicit requested-horizon design is selected, one model query would be

$$Q_{\theta(s_t,e,i,h)} = Q_{h,e}^{\theta(s_t,i)}, \quad 1 \le h \le b_t \le H$$

rather than a value from a separately configured fixed-H model. The maximum H is a dataset contract and later a model/checkpoint contract. In this candidate design, requested residual horizon h is a model input and remaining budget b_t determines whether the query is admissible. In the fixed-H alternative, h is implicit and b_t remains state context. The step index t stays lineage unless a named non-stationarity ablation uses it.

In the requested-horizon candidate design, one candidate row is one geometric query. Candidate pose, candidate–target relation, and candidate-local support are encoded once, and a lightweight horizon embedding turns that candidate into a horizon–candidate query:

$$u_{t,e,h,i} = \mathrm{CrossAttn}_{\theta\left(x_{t,e,i}+\mathrm{Emb}_{h(h)},\{h_e^{\mathrm{tgt}},\Phi_t^{\mathrm{scene}},\boldsymbol{H}_t,b_t\}\right)}$$

followed by one shared per-row value head. This conditioning follows the general principle of a shared value approximator queried by an explicit task variable [Sch+15]. Fixed-H models and separate $Q_1, ..., Q_H$ heads remain competing designs until the source-owner decision is complete.

This candidate could score one h and return $\left[B, N_q\right]$, or vectorize admissible horizons and return $\left[B, L, N_q\right]$. Static encodings would be reused across the horizon axis. Such queries must receive only the causal state available at step t; the learning target, not future scorer input, determines how many rewards the value represents.

The valid-action mask sanitizes candidate rows, gates the masked policy argmax, and gates every supervised or bootstrap maximum. It becomes an attention key mask only in architectures where candidate rows themselves are keys. Target-independent root encodings may be reused across targets, but target-dependent candidate generators require per-target tables or a union table with an explicit target–candidate availability mask.

The value family is defined relative to a frozen state and source protocol. An `S0-pose` value, a privileged `CF-GT` selected-depth value, and a deployable observed-state value are different functions even when their tensors have similar shapes. Likewise, “optimal” means optimal continuation only within the generated finite candidate support, hard-validity contract, represented state, and transition distribution. A pose-only state cannot be promoted to a task-sufficient reconstruction value merely by requesting a longer horizon.

4.5.3 Horizon-recursive offline learning

Batch fitted Q iteration learns a greedy action-value function from a fixed transition collection through successive supervised regression problems; it does not require online interaction [EGW05]. If the requested-horizon design is selected, one candidate lower-horizon recursion is

$$y_t^{(1,e)} = r_t^e$$

and, for $h > 1$,

$$y_t^{(h,e)} = r_t^e + \gamma(1 - d_t) \max_{j:m_{t+1,j}=1} Q_{|(\theta)}(s_{t+1}, e, j, h-1)$$

where the lower-horizon prediction is detached, frozen, or supplied by a delayed target copy. The essential structural rule for this candidate is $Q_h \leftarrow Q_{h-1}$: no horizon value bootstraps from itself. Fixed-horizon TD motivates this recursion and shows that horizon-indexed values can share parameters and be updated in parallel, although a staged $h = 1$ to H schedule remains the clearest initial control [De +20].

The stored evidence gives a particularly strong base case. Every candidate admitted by `q_train_mask` can supervise continuous one-step root-normalized gain. If a selected first action has a successor table with dense one-step labels, then the exact finite-support H=2 target is

$$y_t^{(2,e),\text{ exact}} = r_t^e + \gamma \max_{j:m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

This exact target is a useful recursion check and H=2 control. Whether the production learner is one requested-horizon scorer or a fixed-H family remains open at the source-owner gate.

Double Q is an optional estimator for the learned successor maximum. It uses the online scorer to select

$$j^\star = \operatorname*{argmax}_{j:m_{t+1,j}=1} Q_{\theta(s_{t+1},e,j,h-1)}$$

and a delayed scorer to evaluate that row. This may reduce overestimation from maximizing noisy learned values [HGS15]. It is neither a requirement for offline learning nor the definition of the requested-horizon design candidate. It cannot repair unsupported long-horizon actions, an aliased actor state, or missing selected-observation evidence.

Complete stored chains also permit regression to truncated Monte-Carlo returns. Those targets estimate the continuation of the behavior policy that generated each chain, Q^μ, rather than the greedy finite-support value $Q^\star$ unless that behavior policy is itself the specified target policy. They are therefore useful controls and diagnostics, but they must not be mixed with optimal Bellman targets without naming the estimand.

The objective-design comparison is therefore:

1. dense continuous Q_1 supervision on every candidate admitted by `q_train_mask`;
2. exact selected-action Q_2 supervision as a base-case certification;
3. fixed-H fitted models versus one explicitly horizon-conditioned model for $h = 1, ..., H$;
4. Double-Q selector/evaluator backups as a max-bias ablation;
5. behavior-policy Monte-Carlo return regression as a separate estimand;
6. an uncentred one-step-plus-residual decomposition.

Because dense one-step rows vastly outnumber selected transitions at longer horizons, training and evaluation must report support, loss, calibration, ranking, and selected-action regret separately by horizon under either interface. If requested horizons share one learner, their sampling or weighting must be explicit; an aggregate loss must not allow Q_1 to hide longer-horizon failure.

4.5.4 One-step base and finite-horizon residual

Implementation: exploratory **Evidence: pending**

The residual decomposition is a testable hypothesis. Direct continuous finite-horizon value prediction remains the required control under whichever time-query interface is selected.

Literature: [De +20, CMR19]

Promotion gate: target-specific root-gain calibration and direct-versus-residual ablation across horizons

Development source: docs/literature/tex-src/arXiv-VIN-NBV/sec/3_methods.tex, ordinal RRI paragraph, line 125; docs/contents/theory/candidate_view_dependence.qmd

The hypothesis separates calibrated immediate utility from downstream effects:

$$b_{\psi,i} = f_\psi^{\text{1-step}}\big(s_t^{\text{cf0}}, \boldsymbol{\varphi}_e, q_{t,i}\big), \quad \delta^h_{\theta,t,e,i} = g_\theta\big(\mathcal{I}_{t,e}, q_{t,i}, h\big), \quad Q_{h,\theta,e,i} = b_{\psi,i} + \delta^h_{\theta,t,e,i}$$

The base $b_{\psi,i}$ must predict continuous one-step target root gain

$$\frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_0^e + \varepsilon}$$

in the same additive units as the finite-horizon return. Historical state-relative RRI or an ordinal CORAL score may remain auxiliary ranking and calibration outputs, but they are not interchangeable with this additive base unless an explicit calibrated conversion is learned and validated. The residual captures candidate regeneration, selected-observation state changes, occlusion, support, and the requested residual horizon.

It is not exactly mean-centred within each candidate table because duplicate or unrelated rows would then change absolute value targets. Magnitude regularization may be tested without redefining the value field:

$$Q_{h,\theta,e,i} = \hat{r}^{e}_{\psi}\left(s^{\mathrm{cf0}}_{t}, \boldsymbol{\varphi}_{e}, q_{t,i}\right) + \delta^{h}_{\theta,t,e,i}\left(\mathcal{J}_{t,e}, q_{t,i}\right), \quad \mathcal{L}_{\delta} = \lambda_{\delta}\frac{1}{|\mathcal{A}_t|}\sum_{j\in\mathcal{A}_t}\left(\delta^{h}_{\theta,t,e,j}\right)^2$$

CORAL remains a motivated one-step ranking and calibration interface,

$$p^{\mathrm{CORAL}}_{t,i,k} = \sigma\left(o^{\mathrm{CORAL}}_{t,i,k}\right), \quad k = 0, \ldots, K-2$$

$$\pi^{\mathrm{CORAL}}_{t,i,k} = p^{\mathrm{CORAL}}_{t,i,k-1} - p^{\mathrm{CORAL}}_{t,i,k}, \quad p^{\mathrm{CORAL}}_{t,i,-1} = 1, \quad p^{\mathrm{CORAL}}_{t,i,K-1} = 0$$

$$\hat{r}^{e}_{\psi}\left(s^{\mathrm{cf0}}_{t}, \boldsymbol{\varphi}_{e}, q_{t,i}\right) = \sum_{k=0}^{K-1}\pi^{\mathrm{CORAL}}_{t,i,k}u_k$$

but additive finite-horizon returns are learned in continuous root-gain units. Fixed-H and requested-horizon models, the exact H=2 control, Double-Q ablation, Monte-Carlo control, and residual learner form separate comparisons rather than one implicit architecture.

Research TODO: Compare staged and joint shared-parameter variable-horizon fitted Q against separate per-horizon heads; include dense Q1, exact Q2, Double Q, behavior-return regression, and the uncentred residual after positive oracle-lookahead headroom is established.

Source: variable-horizon learning-target contract

Gate: A0/A1 learning, per-horizon support report, and headroom report

Open decision: Freeze the maximum horizon, horizon-sampling or loss-weighting rule, lower-horizon target-update schedule, and target-network rule in the resolved training manifest. For the residual ablation, also record whether the one-step base is frozen, slow-fine-tuned, or jointly trained.

Source: variable-horizon fitted-value hypothesis

Gate: model-selection protocol freeze

The scientific endpoint is not training loss. For every claimed horizon, selected trajectories are regenerated under the documented candidate and state protocol and re-scored by the same target-specific oracle used for the baselines. The model succeeds only if it recovers a prespecified fraction of positive oracle-lookahead headroom on held-out scenes without violating horizon, mask, provenance, source, or support constraints.

5 Experimental Design

The experiments separate feasibility evidence from policy evidence. Feasibility requires a frozen scene-level population, reproducible target and candidate support, valid replay linkage, and measured oracle throughput. Policy inference additionally requires a held-out scene split, matched acquisition budgets, independent endpoint oracle evaluation, and an implemented actor-visible decision rule. The current train-only bandwidth pilots address only the first category: incomplete runs and resource failures characterize the generation system, but they cannot support comparisons between policies.

5.1 Study Population and Evidence Gates

The source population comprises ASE/ATEK snippet windows from scenes for which the configured GT mesh and object-box table resolve. A frozen manifest assigns entire scenes to train, validation, or test before model selection; no scene may cross these boundaries through another snippet. Each reported run records the manifest hash and the exact counts of scenes, snippets, admitted target tasks, rollout chains, transitions, and retained candidate rows. A capped train-only pilot is therefore a throughput and support probe, not a sample from which held-out policy performance can be estimated.

The present data generator defines oracle target tasks by seeded sampling from geometry-valid GT OBB rows. Its task-coverage report therefore describes the available GT pool, sampled tasks, classes, scenes, and later oracle-evaluation failures. It does not measure proposal matching, IoU ambiguity, projected visibility, or actor-observation support. Those quantities belong to a future observed-target selector required for deployable-input claims. Until that selector exists, GT target geometry may define labels and bounded oracle references but cannot be presented as actor-visible input. The privileged-supervision boundary in Figure 1 applies equally to render-derived evidence: dense GT candidate depth may produce labels, returns, or explicitly named privileged ablations, but it is not a legal actor input. A separate teacher policy, distillation path, or current-belief renderer remains a hypothesis until implemented and evaluated, so none is promoted to a main-text process figure.

The first policy gate is an actor-visible myopic scorer over the same finite candidate table intended for Q_H. The scorer must expose one value per candidate, respect the hard action mask, and be assessed by candidate ranking, calibration, and oracle-rescored selected actions. The existing scene-level VIN scorer is historical substrate; it is not a target-conditioned control until the observed target descriptor is wired into the model and evaluated on a frozen held-out split.

The second policy gate estimates whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only if the preregistered analysis classifies this headroom as meaningful is Q_H evaluated for recovery from offline rollout traces:

$$\eta_Q = \frac{J_e^{(H)}\big(\pi_Q\big) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is measured by matched endpoint oracle evaluation, not predicted values or training loss. If lookahead has no meaningful headroom, the result is scoped to the frozen split, target protocol, candidate generator, horizon, branch factor, and validity regime. If headroom exists but the learned model does not recover it, target observability, action support, replay coverage, reward construction, and model capacity remain separate candidate explanations.

Claim	Primary evidence	Decision rule
Population	scene-split manifest and coverage bundle	Inference is restricted to the frozen held-out scene population.
Task protocol	GT pool, sampled tasks, classes, and oracle failures	Current evidence is oracle-task coverage, not observed-target matching.
Myopic control	ranking, calibration, and oracle-rescored selections	Actor-visible target conditioning must be implemented before comparison.
Planning headroom	Δ_{look} and recovered fraction η_Q	Q_H is evaluated only after a meaningful-headroom gate.

Table 8: Objective-to-evidence matrix.

5.2 Replay Eligibility and Finite-Horizon Learning Gate

The factual rollout tables determine which records may supervise each learning problem. `valid_action_mask` defines actions selectable under the admitted V0 geometry contract. The stricter `q_train_mask` additionally requires a valid target/GT label state and finite target-root-gain and diagnostic target-RRI labels. Padding, actor validity, one-step training eligibility, transition eligibility, modality presence, source role, and horizon availability remain separate masks. Masked rows remain available for support and failure analysis but cannot enter action selection, supervised loss, or bootstrap maximization.

All eligible candidate rows can support dense one-step supervision. Exact H=2 supervision is narrower: the factual first action must have a stored reward and a valid successor step whose candidate table exposes at least one finite one-step root-gain label. General recursive supervision is narrower again: it requires a factual selected action, reward, terminal flag, discount, a defined training horizon, and —when nonterminal—a reproducible successor state and hard mask. The derived `q_h/` arrays align these fields on a padded state–candidate view; they do not create labels for unobserved transitions, make selected GT depth actor-visible, or turn sparse long-horizon action support into dense support.

Learning surface	Purpose	Minimum factual evidence
dense $h = 1$	immediate candidate value	actor-selectable row with finite one-step root-gain label
exact $h = 2$	base-case finite-support value	selected reward plus successor table with at least one finite one-step root-gain label
recursive $h > 1$	variable-horizon fitted value	selected transition, successor actor state and mask, lower-horizon target support, terminal and discount
behavior return	policy-conditioned Monte-Carlo control	complete retained reward prefix and behavior-policy identity

Table 9: Required replay evidence by learning target. Dense immediate labels, exact H=2 targets, recursive optimal-continuation targets, and behavior-policy returns are distinct supervision surfaces.

Gate	Available evidence	Required before inference
Oracle data	target labels, masks, lineage, replay validation, and selected-depth persistence	held-out coverage, source-role audit, and matched endpoint evaluation
Myopic control	scene-level VIN substrate	actor-visible target-conditioned Q_1 scorer and frozen checkpoint
Finite-horizon scorer	selected-transition training seam; no production scorer yet	source-owner interface decision, exact-Q2 control, state-protocol freeze, compatible checkpoint, and oracle-rescored policy
Requested-horizon alternative	selected transitions; explicit horizon-query contract remains proposed	comparison against fixed-H, certified recursion, supported horizons through H, and per-horizon validation
Dynamic Q_H	selected-observation persistence and planned state update	typed dynamic-state reader, deterministic fusion, source masks, and held-out policy evaluation
Policy claim	train-only feasibility pilots	completed held-out paired comparison under equal budget

Table 10: Learning-readiness gates. Scorer-independent training infrastructure does not establish a runnable finite-horizon scorer or policy.

Implementation: partial **Evidence: pending**

A masked selected-transition Double-Q learner for an injected scorer is implemented. No H=2 V0 pose-history scorer, shared requested-horizon scorer, task-sufficient dynamic state, or policy evidence is implemented.

Literature: [De +20, EGW05, FMP19, HGS15, Kum+20]

Promotion gate: fixed-H versus requested-horizon source-owner decision, production scorer, exact Q2 certification, supported H>2 targets, compatible checkpoint, frozen state protocol, and held-out oracle re-evaluation

Development source: aria_nbv/aria_nbv/data_handling/qh.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/rollouts/qh_reader.py

The finite-candidate value model decodes actions only over valid candidate rows:

$$\boldsymbol{u}_{t,i} = \mathrm{Enc}_\theta\big(\mathcal{J}_{t,e}, q_{t,i}\big)$$

$$a_t^\theta = \operatorname*{argmax}_{i:m_{t,i}=1} Q_{h,\theta,e}(s_t, i)$$

The masked argmax is already the discrete decision rule. A separate actor network and online data collection are not required to train or execute this finite-candidate policy. Batch fitted Q iteration explicitly learns a greedy Q function from a fixed collection of transitions by repeatedly solving supervised regression problems [EGW05].

5.2.1 Requested-horizon target candidate

If the source-owner decision selects an explicit requested-horizon interface, one candidate model is $Q_{\theta(s_t,e,i,h)}$ for each residual horizon h admitted by $1 \leq h \leq b_t \leq H$. The boundary target is

$$y_t^{(1,e)} = r_t^e$$

and the recursive target is

$$y_t^{(h,e)} = \begin{cases} r_t^e & \text{if } \;h = 1 \;\text{ or }\; d_t = 1 \\ r_t^e + \gamma \max_{j:m_{t+1,j}=1} Q_{|(\theta)}\big(s_{t+1}, e, j, h-1\big) & \text{if } \;h > 1 \;\text{ and }\; d_t = 0 \end{cases}$$

The lower-horizon prediction is treated as a fixed regression target by stop-gradient, a frozen stage checkpoint, or a delayed target copy. For this candidate, the defining recursion is $Q_h \leftarrow Q_{h-1}$ rather than $Q_h \leftarrow Q_h$. Fixed-horizon TD was introduced precisely for predictions over a bounded number of future rewards and avoids same-horizon self-bootstrapping; its horizon functions may use shared parameters and parallel updates [De +20].

For the requested-horizon candidate, the clearest initial schedule is staged backward induction with one shared horizon-conditioned network:

1. fit Q_1 from dense one-step labels for every candidate admitted by `q_train_mask`;
2. freeze or snapshot the lower-horizon target path;
3. fit Q_2 from selected transitions and validate it against the exact dense-successor target;
4. continue through Q_H, always requesting $h-1$ from the successor target path;
5. optionally fine-tune all horizons jointly after the staged model passes per-horizon regression and ranking gates.

This schedule preserves one inference interface and shared encoders while making target lineage explicit. Fixed-H networks and separate per-horizon heads remain competing designs, not merely controls, until the scorer interface decision is complete.

For remaining horizon two, the store supplies an exact target whenever the successor table has dense one-step labels:

$$y_t^{(2,e),\text{ exact}} = r_t^e + \gamma \max_{j:m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

This target uses no learned successor value or target network. Agreement between fitted Q_2 and this exact control is a required base-case test before interpreting longer-horizon results under either scorer interface.

5.2.2 Double-Q and behavior-return controls

Double Q changes how a noisy learned successor maximum is estimated; it does not decide between fixed-H and requested-horizon scorer interfaces. In the requested-horizon candidate, the online path selects

$$j^\star = \operatorname*{argmax}_{j:m_{t+1,j}=1} Q_{\theta(s_{t+1},e,j,h-1)}$$

and the delayed path evaluates $Q_{|(\theta)}(s_{t+1},e,j^\star,h-1)$. This selector/evaluator split can reduce overestimation caused by maximizing noisy action values [HGS15].

It remains an ablation against the simpler frozen lower-horizon maximum. It is relevant in an offline setting only because a learned maximum is present, not because online learning is planned.

A retained chain also yields the truncated Monte-Carlo target

$$G_{t,e}^{(h),\mu} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e$$

for its behavior policy μ. Regression to this fixed target is a useful policy-conditioned control, but it estimates Q^μ, not the greedy finite-support value $Q^\star$, unless μ is explicitly the target continuation policy. Behavior returns from random-valid, greedy, softmax, and oracle-lookahead chains must therefore remain identified rather than pooled as if they represented one optimal value function.

Double Q addresses one maximization bias. It does not create missing successor transitions, make unsupported actions reliable, or repair state aliasing. CQL and BCQ motivate explicit offline-support diagnostics because a greedy learned policy can select actions whose multi-step consequences are weakly represented in the behavior data [FMP19, Kum+20]. Conservative regularization is introduced only if those diagnostics reveal systematic unsupported-action overestimation.

5.2.3 Horizon-balanced replay and evaluation

Dense one-step rows vastly outnumber selected-action targets at longer horizons. An unweighted row mean can therefore optimize the myopic task while obscuring longer-horizon failure. If one shared requested-horizon learner is selected, the training manifest must freeze either:

- a horizon-stratified sampler;
- per-horizon loss weights w_h;
- or a fixed number of admitted targets per horizon and scene.

Every run reports, separately for each h:

- admitted states, selected actions, behavior policies, candidate families, scenes, and targets;
- value loss and signed target error;

- candidate ranking and top-action regret where oracle comparison is available;
- bootstrap, terminal, and no-valid-successor fractions;
- online/target disagreement for Double-Q runs;
- endpoint performance of the masked policy requested at the remaining budget.

A single scalar validation loss is insufficient for model selection unless its horizon aggregation is frozen in advance. Cross-stage corpus admission must also require the same maximum horizon, reward and return semantics, discount, state/source protocol, candidate/reason vocabulary, and horizon-weighting rule.

The optimality claim remains bounded under either interface. A learned finite-horizon value can approximate the best continuation only within the sampled finite candidate generator, hard-validity regime, represented actor state, and offline transition support. The same checkpoint cannot silently mix an `S0-pose` state with `CF-GT`, sensor-like, or V1 dynamic states. Longer horizons increase—not decrease—the need for selected-observation geometry and a sufficiently Markov scene state.

The current scorer-independent learner establishes infrastructure readiness only. Confirmatory interpretation requires the source-owner scorer decision, a canonical V0 corpus, dense-Q1 and exact-Q2 controls, supported targets for every claimed horizon, cross-stage learning-contract equality, checkpoint compatibility, a frozen actor-state protocol, horizon-specific support diagnostics, and matched held-out endpoint oracle re-evaluation. A dynamic-state claim additionally requires selected-observation fusion and no-future-observation leakage tests.

5.3 Policy Comparison and Statistical Protocol

Implementation: partial **Evidence: pending**

The aggregation and artifact-driven reporting seam exists, but confirmatory paired policy evidence does not.

Promotion gate: frozen held-out manifest, completed paired endpoints, and validated report bundle

Development source: aria_nbv/aria_nbv/rollouts/reporting.py; docs/typst/thesis/experiment_data.typ

The experimental unit is the scene. Every comparison is paired by scene, target task, root state, candidate-generation distribution, hard validity regime, and acquisition horizon. Planner depth may differ because it defines the decision rule, but every policy receives the same acquisition budget. Repeated snippets, targets, and seeds are aggregated within scene before the primary comparison so that densely sampled scenes do not dominate the estimand.

The primary estimand is the paired per-scene difference in fixed-budget endpoint target-quality gain. The centralized analysis artifact freezes the scene aggregation function, uncertainty procedure, interval level, comparison set, and minimum meaningful effect before final aggregation. Secondary summaries—cumulative root-normalized gain, invalid-action rate, runtime, path length, and task/candidate coverage—diagnose mechanism and feasibility but do not replace the endpoint estimand. The current replay store alone cannot estimate this endpoint comparison because it does not persist an independent post-budget reconstruction for every policy; confirmatory analysis therefore requires matched endpoint oracle re-evaluation or an equivalent frozen endpoint record.

Support and failure strata are fixed before policy inspection. The report distinguishes source rows without an admitted target task, admitted tasks whose oracle evaluation fails, roots without a valid action, trajectories that terminate before the shared budget, and completed paired trajectories. A policy that cannot continue remains in the comparison with no further gain; it is not silently dropped. These strata delimit the analysis population and expose whether a policy difference is instead a support or systems failure.

Policy	Decision information	Acquisition budget	Scientific role
π_{rand}	actor-visible	H	valid-action lower reference
$\pi_{\text{learned-1}}$	actor-visible	H	learned myopic control
$\pi_{\text{oracle-1}}$	oracle immediate reward	H	one-step oracle-greedy comparator
$\pi_{\text{oracle-look}}$	bounded oracle lookahead	H	conditional finite-support upper reference
π_Q	actor-visible	H	planned learned finite-horizon recovery

Table 11: Matched policy comparison. All rows acquire the same number of views; oracle access and planner depth define the decision rule rather than the acquisition budget.

The first inferential comparison is bounded oracle lookahead against one-step oracle greedy. Repeated oracle evaluation estimates the metric noise floor, and the analysis artifact freezes the meaningful-headroom rule before learned-policy inspection. Recovered-headroom ratios are reported only when their denominator passes that gate. The learned comparison then contrasts Q_H with the actor-visible myopic control under the same endpoint evaluation and acquisition budget; because neither learned target-conditioned control is presently complete, this comparison remains prospective.

Validation TODO: Populate the evidence chain in order: metric repeatability, candidate and target support, oracle-lookahead headroom, myopic-control calibration, finite-horizon recovery, then representation and architecture ablations. Missing upstream evidence blocks downstream claims rather than becoming a zero result.

Source: thesis objective-to-evidence contract

Gate: artifact-backed Results bundle

Validation TODO: Run row-shuffle, mask-isolation, duplicate-row, valid-count, frame-transform, target-source-dropout, and horizon-boundary tests for every model admitted to the policy comparison.

Source: geometric acceptance contract

Gate: architecture validity report

Open decision: Freeze the scene aggregation, interval procedure, interval level, comparison family, and meaningful-headroom threshold in the resolved analysis manifest rather than in thesis prose.

Source: artifact-driven reporting plan

Gate: confirmatory analysis freeze

5.4 Outcome Logic

Meaningful, stable lookahead headroom establishes only that the frozen finite-candidate setup contains exploitable non-myopic structure. No meaningful headroom is a setup-specific negative result, not evidence that target-aware planning is universally myopic. An unstable oracle metric or insufficient paired support blocks downstream policy claims. Likewise, train-only bandwidth pilots and failed generation attempts, including memory-exhaustion failures, inform compute configuration and throughput planning only; incomplete artifacts cannot establish headroom or policy superiority.

6 Results

The thesis report bundle is loaded through the strict schema checked in `experiment_data.typ`. Its declared evidence class is *pilot*. Schema validity establishes only that provenance, missingness, and table contracts are readable; it does not turn a development fixture or an incomplete pilot into scientific evidence.

Result	**Required evidence**	**Status**
Population and targets	population facts with denominators in every validated store	not available
Candidate support	valid/total candidates and zero-action failures in every validated store	not available
Paired policy effect	effect, interval, scene count, and headroom gate in every validated store	not available
Resource feasibility	wall time, peak GPU memory, and storage in every validated store	not available

Table 12: Evidence availability in the loaded report bundle. “Not available” means that no thesis result is inferred from fixture values, train-only attempts, or an incomplete store.

6.1 Candidate and Store Feasibility

The available artifacts show that the finite-candidate rollout path reaches mesh rendering, target-specific oracle scoring, and selected-action replay on training sources. They therefore support an implementation-readiness claim only. A CUDA out-of-memory failure in an unbatched candidate render and later memory-bounded attempts identify rendering as a scale gate; neither establishes rollout throughput, storage cost, candidate-family support, or policy quality for the intended study population.

6.2 Target-Task Coverage

The implemented pipeline samples geometry-valid GT target tasks and records target and validity fields in the rollout schema. The loaded bundle does not contain a validated population and exclusion table for the study. Scene coverage, eligible-target coverage, invalid-reason frequencies, and their denominators are consequently unavailable as results.

6.3 Policy Comparison

The primary estimand is not estimable from the current evidence because no validated held-out bundle contains the matched policy outcomes and aggregation inputs. Oracle repeatability, oracle-lookahead headroom, learned one-step performance, finite-horizon recovery, uncertainty intervals, and sensitivity analyses are therefore not reported.

6.4 Runtime and Storage Gate

The runtime and storage claim requires completed validated stores whose resolved manifests, failure records, and resource summaries refer to the same run. Until that evidence exists for both rollout profiles, the large-scale data-generation gate remains pending. The current attempts justify memory-bounded rendering and retained failure provenance, but no extrapolation to cluster throughput or dataset volume.

7 Discussion

The current evidence supports a narrow implementation conclusion. ARIA-NBV has an executable finite-candidate path that separates actor-visible policy inputs from oracle-only target geometry, applies invalidity as a hard decision constraint, evaluates target-specific reconstruction change offline, and exposes the resulting store through one typed reporting seam. This establishes that the proposed experiment can be represented and audited; it does not establish that one candidate family, rollout policy, representation, or learned model performs better than another.

The rollout attempts also expose a systems limitation. Counterfactual scoring repeatedly renders a large mesh for a candidate set, so renderer memory and latency constrain feasible branch factor and rollout volume before statistical efficiency becomes relevant. An out-of-memory observation is evidence for batching and resource measurement, not for or against the candidate distribution or planning objective. Completed validated stores are required before throughput, storage, or failure rates are generalized.

The scientific limitations are more restrictive. Current attempts use training sources, no held-out paired policy table supports the endpoint estimand, metric repeatability and meaningful oracle-lookahead headroom remain unestablished, and the target-conditioned finite-horizon comparison is not implemented. The present evidence therefore cannot distinguish negligible non-myopic structure from inadequate candidate support, target observability, replay coverage, representation support, or model failure.

7.1 Evidence-conditioned architecture bridges

Implementation: exploratory **Evidence: pending**

Alternative representations and policies remain part of the thesis design registry, but they are promoted only as responses to diagnosed bottlenecks.

Promotion gate: a measured limitation that the proposed bridge directly tests

Development source: Table 4; Table 7

If the local EFM3D field is the limiting variable, broad semidense memory, sparse occupied/free/unknown state, target-centred re-lifting, or logged appearance features provide progressively stronger representation tests. If compact descriptors lose relevant spatial structure, point, sparse, or equivariant encoders become justified. If independent candidate queries leave errors correlated with the valid candidate set, DeepSets or masked candidate interaction becomes relevant. If selected-view history fails specifically by approach direction, signed first- plus second-moment memory, followed only if needed by spherical harmonics, becomes the targeted ablation. A renderable 3DGS state is appropriate only when candidate rendering or soft instance membership is itself the measured missing capability.

Continuous and hierarchical policies change the action contract rather than merely increasing model capacity. A target-then-pose policy could first select or mask a target token and then condition pose queries on target, horizon, and step; multiple targets can share scene encodings and be evaluated in parallel. This remains post-core work because continuous pose generation requires separate feasibility, safety, support, and simulator-realism evidence. The finite-candidate model is retained as the calibrated control even if a continuous policy is later introduced.

Sequence or recurrent models are similarly conditional. They become scientifically motivated when errors persist after explicit history, time, horizon, and scene-memory conditioning, not simply because the dataset contains trajectories. Looping or recurrent refinement must preserve causal masks and candidate-row equivariance and must be compared against the same fixed-budget endpoint oracle evaluation.

Research TODO: After the primary evidence chain is complete, promote only bridges tied to a measured failure: EVL support, target observability, directional history, valid-set interaction, long-horizon credit, or action-support mismatch.

Source: method design-space registry

Gate: artifact-backed failure analysis

The next evidential step remains procedural: complete validated stores under resolved manifests, freeze the eligible population and exclusions, establish oracle stability and headroom, train the matched controls, and only then interpret architectural differences. The registry above preserves concrete follow-up hypotheses without presenting them as validated conclusions.

8 Conclusion

This thesis defines a leakage-auditable experiment for target-conditioned finite-candidate next-best-view planning. Its present contribution is the separation of actor-visible state from oracle supervision, the target-specific reconstruction objective, hard validity and replay contracts, and an artifact-driven reporting seam that keeps provenance and missingness attached to later results.

The available evidence does not answer whether bounded oracle lookahead improves fixed-budget target reconstruction or whether a learned finite-horizon policy recovers such headroom. The current training-source rollout attempts establish pipeline reachability and reveal a renderer resource gate; the development report fixture establishes the data contract only. Neither supports a held-out policy claim, a population-level effect, or a scale estimate.

The final scientific conclusion is therefore conditional on evidence that is not yet available. A stable oracle metric and positive paired lookahead effect would define measurable headroom for the evaluated finite support. Negligible headroom would be a setup-specific negative result. Unstable oracle evaluation would block planning claims, and stable headroom without learned recovery would remain a learned-control failure with several unresolved mechanisms. These outcomes delimit the thesis without extending it to continuous control, online reinforcement learning, or real-device deployment.

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A Reproducibility Record

The report bundle is the single numerical interface between validated rollout artifacts and this manuscript. Its schema version, evidence status, store manifests, typed parameter rows, failure records, and sidecar hashes preserve provenance without duplicating analysis logic in Typst. The document loader rejects schema or status mismatches before any table is rendered. Compact labels and digest prefixes are used below; complete store identifiers, paths, and hashes remain in the machine-readable bundle.

Evidence class	**Store**	**Manifest digest**	**Validated**
Development fixture	store S1	not-a-real-m…	yes

Table 13: Loaded development-fixture provenance. The fixture verifies the document interface only and is not scientific evidence.

The bundled development fixture contains synthetic values solely to test typed loading and missingness. Its numerical rows are deliberately not rendered or interpreted.

B Development Diary

Archived source note: This marked diary is a development-only record. It is excluded from submission mode; only evidence-backed definitions, protocols, and results graduate into the thesis body.
Source: thesis mode contract

B.1 Decisions Retained

The thesis keeps the privileged oracle and actor-visible policy as separate contracts, evaluates finite candidate sets under a shared validity mask, and treats target-specific reconstruction improvement as the common comparison signal. Rollout stores retain selected transitions and provenance rather than becoming a second implementation of the oracle or training pipeline.

Open decision: Freeze the scene-disjoint analysis population and the minimum evidence scale only when the resolved manifests and exclusion ledger are available.
Source: experimental-design contract
Gate: confirmatory bundle freeze

B.2 Failures That Changed the Plan

Early rollout attempts reached the mesh-rendering path but exceeded available accelerator memory when candidate views were rendered without a bounded batch. This failure narrowed the immediate claim to implementation readiness and made peak memory, throughput, failure provenance, and completed-store validation explicit scale gates.

Validation TODO: Require completed stores, matching manifests, resource summaries, and recorded failures before extrapolating rollout bandwidth or storage demand.
Source: rollout attempt artifacts
Gate: cluster-readiness evidence

B.3 Rejected Scope

Continuous actions, online simulators, additional scene representations, and alternative reinforcement-learning families are excluded from the core study while the finite-candidate target-conditioned comparison remains unvalidated. They would change both the action contract and the supervision regime, so adding them now would weaken rather than extend the primary experiment.

Remove or rewrite before submission: Delete bridge material that cannot be tied to a concrete limitation or a matched follow-up experiment after the confirmatory analysis is complete.

Source: scope review

Gate: discussion freeze

B.4 Directional-Memory Hypothesis

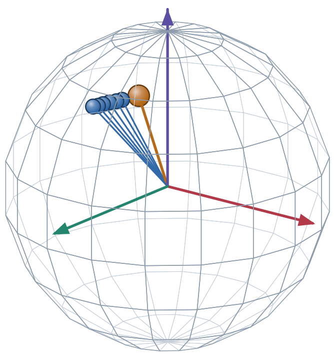
Research TODO: Treat target-centred directional memory as an ablation hypothesis, not an actor feature or contribution. The representation must be implemented, mask-audited, and compared against pose distance and overlap before any predictive claim enters the manuscript.

Source: RQ3 representation hypothesis

Gate: paired held-out ablation

A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

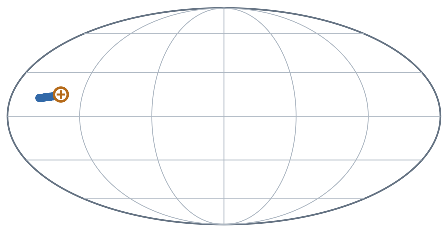


● logged history direction ⊕ example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



$+x_e$ at map centre · $+z_e$ north · wrap seam at ±180°

$$\boldsymbol{M}_{\mathrm{dir}(v)} = \sum_{k<t} w_{k(v)} \boldsymbol{d}_{k(v)} \boldsymbol{d}_{k(v)}^{\top}$$

$$\nu_{i(v)} = 1 - \frac{\boldsymbol{d}_i^{\top} \boldsymbol{M}_{\mathrm{dir}} \boldsymbol{d}_i}{\mathrm{tr}\, \boldsymbol{M}_{\mathrm{dir}} + \varepsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\boldsymbol{M}_{\mathrm{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.

Figure 7: Development-only directional-memory fixture. The sphere and Mollweide panels show the same logged ASE camera directions in a target-object frame; the orange query is another logged pose used to inspect one prospective second-moment novelty definition. No smooth field, spherical-harmonic representation, or learned benefit is claimed.

B.5 Next Evidence Gates

The next admissible evidence is a validated report bundle that fixes the study population, records target and candidate exclusions, supplies paired endpoint outcomes with uncertainty, and joins runtime and storage statistics to the same manifests. Only then can the manuscript replace the present availability statements with policy comparisons.

Implementation TODO: Export the confirmatory report bundle, compile both thesis modes against it, and inspect the resulting tables and failure cases before freezing claims.

Source: reporting contract

Gate: claim freeze

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbststaendig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Muenchen, 7. Jan 2027 Jan Duchscherer